perotac
MADE IN TAIWAN

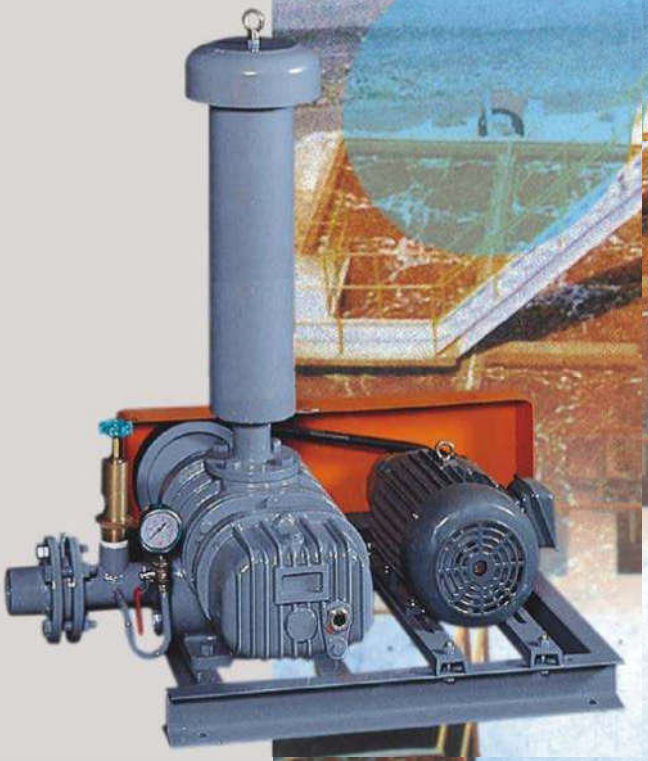
# PEROTAC

## Roots Blowers
## Membrane Disc Diffuser

ISO 9001
BSMI

CE

ISO-9001/CNS12681





# THREE LOBES ROOTS BLOWER

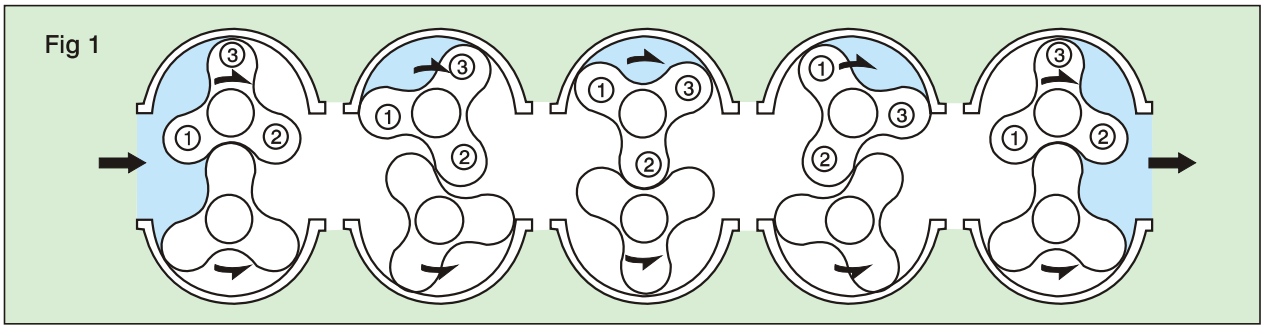
## GENERAL

PEROTAC 3 LOBES ROOTS BLOWERS, Know also as POSITIVE DISPLACEMENT ROTARY LOBE PUMPS, are used in conditions ranging from strong vacuum to high pressure in all branches of industry, especially in waste water treatment plant. The simple design, easy handling, and stable performance make possible a wide range of applications. Their main characteristic is to produce a constant flow of gas at different pressure and efficiencies are normally high.
Another important feature is that the pumped gas does not get contaminated because there are no lubricants in the area in which the compression occurs.
The blowers consist of a cast-iron casing in which two rotors with conjugate shape rotate and are sync-hronized by a set of timing gears. During the rotation, there is no contact either between the lobes or between the lobes and the casing.
Two front and rear side covers close each casing : the bearing of the shafts and the sealing devices to avoid gas leakage are located there.

## OPERATING PRINCIPLE

The two three-lobe rotors rotate one opposite to the other; their movement is synchronized by a set of timing gears.
The geometric configuration of the lobes allows them to have, at any moment, a generatrix in common; due to very low machine tolerance, this common generatrix acts also as a seal device for both rotors. During the operation, one or two lobes of each rotor come into contact with the internal surface of the casing : this create a chamber in which the gas is trapped.
As the rotation continues, the trapped gas moves along until it reaches the position of discharging port. The two rotors make six intake and exhaust cycles per revolution; therefore the capacity of the PEROTAC roots blower is determined by its operating speed and increase proportionally with the speed, but is independent from the pressure differential between inlet and outlet ports.

Fig 1

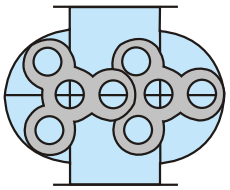



## ADVANTAGES OF 3 LOBES DESIGN

Owing to new 3 lobes' rotor design, the reverse flow pressure variation period is only 2/3 of that of a conventional 2 lobes' rotor, the peak pressure value is also lower. So the noise and pressure pulsation are greatly limited.
Advantages of PEROTAC 3 LOBES ROOTS BLOWER are as follows :

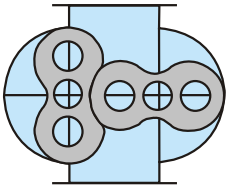
1. Less back flow, steady air flow rate, lower vibration and noise.
2. Discharge pressure pulsation are reduced, so lighter loading of bearings and timing gears ensures the long servicing life.
3. Running on the same operating speed, the 3 lobes blower can deliver large air flow and volume than conventional 2 lobes type.
4. Adequate clearance between the rotors and the rotors with the casing ensure no contact during operation, and this make the efficiency even higher.
5. The precision of rotors is fully controlled and variation of precision between blowers is almost nil because the rotors are produced by utilizing a precision NC machine.
6. The rotors are dynamically balanced in the fabrication stage already, so these rotors are almost free from vibrations ◦

**3-Lobes Cylinder**



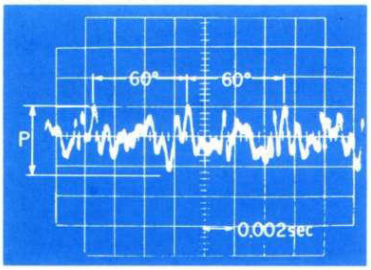
(Fig.1)

**2-Lobes Cylinder**

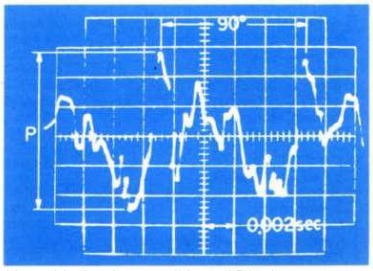


(Fig.2)

**Discharge Pressure Variation in a 3-Lobes Rotor**




**Discharge Pressure Variation in a 2-Lobes Rotor**




Note: Under the condition of Discharge Pressure at 5000 mmAq and Revolution of 1560 r.p.m. for both blowers.

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# CHARACTERISTICS

RL



RW



RECOMMENDED PRESSURE OPERATION RANGE :
RL: 1000~8000mmAq RW: 8000~10000mmAq

## ■ MAXIMUM PERFORMANCE

1 atm = 10333mmAq = 760 mmHg

| | SINGLE STAGE | | DOUBLE STAGE | |
|---|---|---|---|---|
| TYPE | RL | RW | RL | RW |
| MAXI. PRESSURE | 10,000mmAq ( 1.0kg/cm$^2$ ) | 12,000mmAq ( 1.2kg/cm$^2$ ) | 15,000mmAq ( 1.5kg/cm$^2$ ) | 20,000mmAq ( 2.0kg/cm$^2$ ) |
| MAXI. VACUUM | -6,700mmAq ( -500mmHg ) | -8,100mmAq ( -600mmHg ) | -8,100mmAq ( -600mmHg ) | -8,800mmAq ( -650mmHg ) |

## ■ STRUCTURE ANALYSIS

| | GREASE LUBRICATION | OIL LUBRICATION | WATER JACKET FOR COOLING |
|---|---|---|---|
| FRONT BEARING | RW | RL | RW |
| REAR BEARING | – | RL、RW | RW |
| GEAR BOX | – | RL、RW | RW |

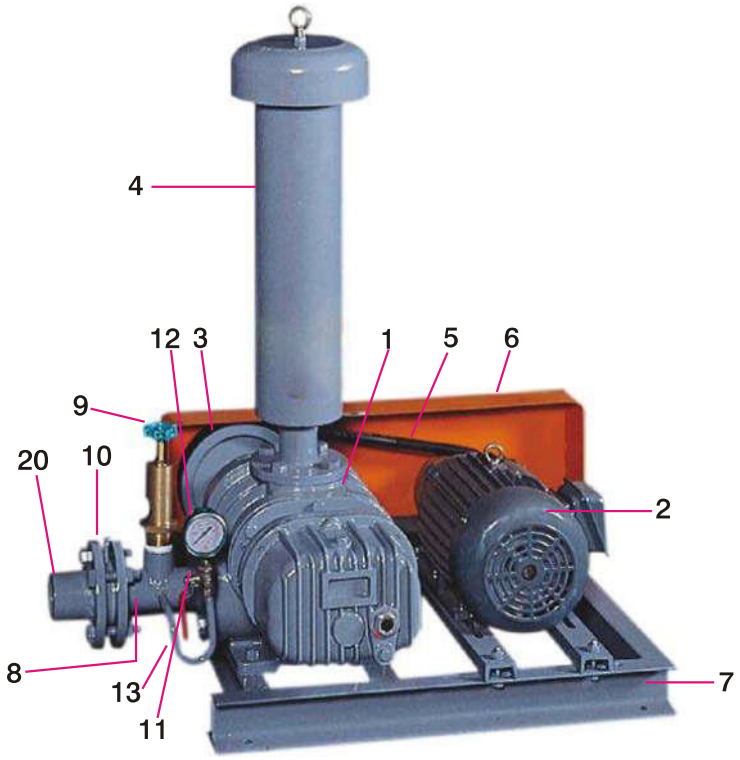
# SPECIFICATIONS

**RL / RW** : Blower Body ONLY.
**RLC/RWC** : Complete Set. - Blower Body With Standard Accessories For Pressure Operation.
**RLV/ RWV** : Complete Set. - Blower Body With Standard Accessories For Vacuum Operation.
Please refer to page 3 for the picture of complete set.

| TYPE | BORE (mm) | BLOWER BODY | COMPLETE SET FOR PRESSURE OPERATION | COMPLETE SET FOR VACUUM OPERATION |
|---|---|---|---|---|
| RL | 40 | RL-40 | RLC-40 | – |
| | 50 | RL-50 | RLC-50 | RLV-50 |
| | 65 | RL-65 | RLC-65 | RLV-65 |
| | 80 | RL-80 | RLC-80 | RLV-80 |
| | 100 | RL-100 | RLC-100 | RLV-100 |
| | 125 | RL-125 | RLC-125 | RLV-125 |
| | 150 | RL-150 | RLC-150 | RLV-150 |
| | 200 | RL-200 | RLC-200 | RLV-200 |
| | 250 | RL-250 | RLC-250 | RLV-250 |
| | 300 | RL-300 | RLC-300 | RLV-300 |
| RW | 50 | RW-50 | RWC-50 | RWV -50 |
| | 65 | RW-65 | RWC-65 | RWV-65 |
| | 80 | RW-80 | RWC-80 | RWV-80 |
| | 100 | RW-100 | RWC-100 | RWV-100 |
| | 125 | RW-125 | RWC-125 | RWV-125 |
| | 150 | RW-150 | RWC-150 | RWV-150 |
| | 200 | RW-200 | RWC-200 | RWV-200 |
| | 250 | RW-250 | RWC-250 | RWV-250 |
| | 300 | RW-300 | RWC-300 | RWV-300 |

## COMPLETE SET ROOTS BLOWER INSTALLATION : with standard accessories

### ■ PRESSURE OPERATION ( RLC 、RWC )




## STANDARD ACCESSORIES

| NO | NAME |
|---|---|
| 1 | BLOWER BODY |
| 2 | MOTOR |
| 3 | BLOWER PULLEY |
| 4 | SUCTION SILENCER SET (WITH AIRFILTER & COVER) |
| 5 | V-BELT |
| 6 | BELT COVER |
| 7 | BASE PLATE |
| 8 | T-TYPE REDUCER |
| 9 | PRESSURE SAFETY VALVE |
| 10 | CHECK VALVE |
| 11 | BALL VALVE |
| 12 | PRESSURE GAUGE |
| 13 | GAUGE PIPE |
| 20 | DISCHARGE FLANGE (OPTIONAL) |

# RL PERFORMANCE TABLE (FOR PRESSURE OPERATION)

■ RLC - Complete Set ■ RL - Blower Body only

| TYPE | Bore | Revolution | 1000mmAq | | 2000mmAq | | 3000mmAq | | 4000mmAq | | 5000mmAq | | 6000mmAq | | 7000mmAq | | 8000mmAq | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 9.8kpa | | 19.6kpa | | 29.4kpa | | 39.4kpa | | 49.0kpa | | 58.8kpa | | 68kpa | | 78.4kpa | |
| | Inch(mm) | R.P.M. | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw |
| 40 | 1½(40) | 800 | 0.70 | 0.25 | 0.56 | 0.50 | 0.45 | 0.77 | 0.35 | 0.94 | 0.26 | 1.14 | 0.21 | 1.34 | 0.15 | 1.52 | 0.10 | 1.70 |
| | | 900 | 0.85 | 0.28 | 0.71 | 0.56 | 0.61 | 0.81 | 0.50 | 1.05 | 0.42 | 1.27 | 0.36 | 1.50 | 0.30 | 1.71 | 0.26 | 1.91 |
| | | 1000 | 1.01 | 0.31 | 0.87 | 0.62 | 0.75 | 0.90 | 0.67 | 1.16 | 0.58 | 1.43 | 0.50 | 1.67 | 0.46 | 1.90 | 0.40 | 2.12 |
| | | 1100 | 1.17 | 0.35 | 1.03 | 0.68 | 0.92 | 0.99 | 0.82 | 1.34 | 0.73 | 1.57 | 0.67 | 1.84 | 0.61 | 2.09 | 0.56 | 2.34 |
| | | 1200 | 1.32 | 0.36 | 1.19 | 0.74 | 1.08 | 1.08 | 0.98 | 1.40 | 0.88 | 1.71 | 0.83 | 2.01 | 0.77 | 2.29 | 0.72 | 2.45 |
| | | 1300 | 1.48 | 0.41 | 1.34 | 0.80 | 1.23 | 1.17 | 1.14 | 1.52 | 1.05 | 1.86 | 0.98 | 2.18 | 0.93 | 2.47 | 0.88 | 2.56 |
| | | 1400 | 1.64 | 0.44 | 1.51 | 0.86 | 1.40 | 1.26 | 1.29 | 1.63 | 1.22 | 1.99 | 1.14 | 2.34 | 1.08 | 2.66 | 1.04 | 2.66 |
| | | 1500 | 1.80 | 0.47 | 1.66 | 0.92 | 1.56 | 1.35 | 1.45 | 1.75 | 1.36 | 2.14 | 1.30 | 2.51 | 1.24 | 2.85 | 1.20 | 2.98 |
| | | 1600 | 1.96 | 0.50 | 1.82 | 0.98 | 1.72 | 1.44 | 1.61 | 1.87 | 1.52 | 2.28 | 1.46 | 2.67 | 1.41 | 3.04 | 1.36 | 3.30 |
| | | 1700 | 2.13 | 0.57 | 1.98 | 1.05 | 1.87 | 1.53 | 1.77 | 1.99 | 1.68 | 2.42 | 1.61 | 2.84 | 1.55 | 3.23 | 1.52 | 3.67 |
| | | 1800 | 2.29 | 0.57 | 2.14 | 1.11 | 2.03 | 1.62 | 1.93 | 2.11 | 1.83 | 2.56 | 1.77 | 3.01 | 1.71 | 3.42 | 1.67 | 3.93 |
| | | 1900 | 2.45 | 0.60 | 2.30 | 1.20 | 2.19 | 1.71 | 2.09 | 2.23 | 1.99 | 2.70 | 1.92 | 3.18 | 1.86 | 3.61 | 1.82 | 4.26 |
| 50 | 2(50) | 800 | 1.16 | 0.42 | 0.92 | 0.83 | 0.75 | 1.29 | 0.58 | 1.56 | 0.44 | 1.90 | 0.34 | 2.22 | 0.24 | 2.53 | 0.17 | 2.83 |
| | | 900 | 1.42 | 0.47 | 1.18 | 0.93 | 1.00 | 1.35 | 0.84 | 1.75 | 0.70 | 2.12 | 0.60 | 2.50 | 0.50 | 2.85 | 0.43 | 3.19 |
| | | 1000 | 1.68 | 0.52 | 1.45 | 1.03 | 1.26 | 1.50 | 1.10 | 1.94 | 0.96 | 2.38 | 0.85 | 2.78 | 0.76 | 3.17 | 0.68 | 3.54 |
| | | 1100 | 1.94 | 0.58 | 1.72 | 1.13 | 1.53 | 1.65 | 1.37 | 2.24 | 1.21 | 2.62 | 1.11 | 3.06 | 1.02 | 3.49 | 0.94 | 3.90 |
| | | 1200 | 2.21 | 0.63 | 1.99 | 1.23 | 1.80 | 1.79 | 1.63 | 2.33 | 1.48 | 2.85 | 1.37 | 3.34 | 1.29 | 3.81 | 1.20 | 4.08 |
| | | 1300 | 2.47 | 0.69 | 2.26 | 1.33 | 2.06 | 1.95 | 1.90 | 2.53 | 1.74 | 3.09 | 1.64 | 3.62 | 1.55 | 4.11 | 1.46 | 4.27 |
| | | 1400 | 2.73 | 0.74 | 2.51 | 1.44 | 2.33 | 2.10 | 2.15 | 2.72 | 2.01 | 3.32 | 1.90 | 3.89 | 1.81 | 4.44 | 1.74 | 4.44 |
| | | 1500 | 3.00 | 0.79 | 2.78 | 1.54 | 2.59 | 2.25 | 2.42 | 2.92 | 2.27 | 3.56 | 2.17 | 4.17 | 2.07 | 4.76 | 2.00 | 4.96 |
| | | 1600 | 3.27 | 0.84 | 3.04 | 1.64 | 2.86 | 2.40 | 2.69 | 3.12 | 2.54 | 3.80 | 2.43 | 4.45 | 2.34 | 5.08 | 2.27 | 5.46 |
| | | 1700 | 3.55 | 0.90 | 3.30 | 1.75 | 3.12 | 2.55 | 2.94 | 3.32 | 2.80 | 4.03 | 2.69 | 4.73 | 2.59 | 5.39 | 2.53 | 6.12 |
| | | 1800 | 3.81 | 0.95 | 3.57 | 1.85 | 3.39 | 2.70 | 3.21 | 3.52 | 3.06 | 4.26 | 2.95 | 5.01 | 2.85 | 5.70 | 2.78 | 6.56 |
| | | 1900 | 4.08 | 1.00 | 3.84 | 1.95 | 3.65 | 2.85 | 3.48 | 3.72 | 3.31 | 4.49 | 3.21 | 5.29 | 3.11 | 6.01 | 3.04 | 7.09 |
| 65 | 2½(65) | 800 | 1.95 | 0.58 | 1.78 | 1.14 | 1.61 | 1.66 | 1.46 | 2.16 | 1.33 | 2.62 | 1.21 | 3.07 | 1.12 | 3.49 | 1.03 | 3.91 |
| | | 900 | 2.28 | 0.79 | 2.11 | 1.28 | 1.94 | 1.86 | 1.79 | 2.42 | 1.65 | 2.95 | 1.54 | 3.45 | 1.45 | 3.93 | 1.35 | 4.40 |
| | | 1000 | 2.63 | 1.14 | 2.46 | 1.42 | 2.29 | 2.06 | 2.12 | 2.69 | 1.99 | 3.28 | 1.89 | 3.83 | 1.78 | 4.35 | 1.69 | 4.89 |
| | | 1100 | 2.97 | 1.29 | 2.80 | 1.57 | 2.62 | 2.28 | 2.47 | 2.96 | 2.34 | 3.61 | 2.23 | 4.22 | 2.12 | 4.81 | 2.03 | 5.38 |
| | | 1200 | 3.27 | 1.44 | 3.14 | 1.71 | 2.96 | 2.49 | 2.80 | 3.23 | 2.68 | 3.93 | 2.56 | 4.60 | 2.46 | 5.24 | 2.36 | 5.86 |
| | | 1300 | 3.63 | 1.54 | 3.48 | 1.85 | 3.29 | 2.70 | 3.14 | 3.50 | 3.01 | 4.26 | 2.90 | 4.98 | 2.80 | 5.68 | 2.70 | 6.35 |
| | | 1400 | 3.99 | 1.64 | 3.82 | 1.99 | 3.64 | 2.91 | 3.50 | 3.77 | 3.36 | 4.59 | 3.24 | 5.37 | 3.15 | 6.12 | 3.04 | 6.84 |
| | | 1500 | 4.34 | 1.74 | 4.16 | 2.13 | 3.98 | 3.11 | 3.83 | 4.04 | 3.70 | 4.91 | 3.58 | 5.75 | 3.48 | 6.55 | 3.38 | 7.32 |
| | | 1600 | 4.67 | 1.84 | 4.51 | 2.27 | 4.32 | 3.32 | 4.17 | 4.31 | 4.03 | 5.24 | 3.92 | 6.13 | 3.82 | 6.98 | 3.72 | 8.28 |
| | | 1700 | 5.02 | 1.94 | 4.86 | 2.42 | 4.66 | 3.53 | 4.52 | 4.57 | 4.37 | 5.57 | 4.26 | 6.51 | 4.17 | 7.42 | 4.07 | 8.30 |
| | | 1800 | 5.36 | 2.04 | 5.20 | 2.56 | 5.00 | 3.73 | 4.86 | 4.85 | 4.71 | 5.89 | 4.60 | 6.89 | 4.50 | 7.85 | 4.41 | 8.78 |
| | | 1900 | 5.71 | 2.14 | 5.53 | 2.70 | 5.34 | 3.94 | 5.19 | 5.12 | 5.05 | 6.22 | 4.94 | 7.27 | 4.84 | 8.28 | 4.75 | 9.27 |
| 80 | 3(80) | 800 | 2.91 | 1.40 | 2.55 | 2.12 | 2.23 | 2.81 | 1.93 | 3.51 | 1.67 | 4.31 | 1.42 | 5.43 | 1.21 | 6.55 | 1.02 | 7.67 |
| | | 900 | 3.51 | 1.50 | 3.25 | 2.32 | 2.93 | 3.14 | 2.63 | 3.84 | 2.36 | 4.64 | 2.12 | 5.76 | 1.89 | 6.88 | 1.70 | 8.00 |
| | | 1000 | 4.21 | 1.60 | 3.95 | 2.53 | 3.63 | 3.47 | 3.33 | 4.25 | 3.06 | 5.52 | 2.82 | 6.64 | 2.61 | 7.06 | 2.42 | 8.18 |
| | | 1100 | 4.91 | 1.70 | 4.65 | 2.75 | 4.33 | 3.80 | 4.03 | 4.70 | 3.76 | 5.71 | 3.52 | 6.83 | 3.31 | 7.95 | 3.12 | 9.07 |
| | | 1200 | 5.40 | 1.80 | 5.14 | 2.91 | 4.82 | 4.03 | 4.52 | 5.04 | 4.29 | 6.05 | 4.07 | 7.17 | 3.84 | 8.29 | 3.65 | 9.41 |
| | | 1300 | 6.05 | 1.90 | 5.75 | 3.20 | 5.43 | 4.25 | 5.13 | 5.60 | 4.95 | 6.72 | 4.69 | 7.84 | 4.48 | 8.96 | 4.29 | 10.08 |
| | | 1400 | 6.53 | 2.00 | 6.27 | 3.47 | 5.95 | 4.59 | 5.65 | 5.82 | 5.38 | 7.05 | 5.14 | 8.28 | 4.93 | 9.56 | 4.74 | 10.81 |
| | | 1500 | 7.00 | 2.10 | 6.74 | 3.50 | 6.42 | 5.04 | 6.12 | 6.27 | 5.85 | 7.50 | 5.61 | 8.96 | 5.40 | 10.42 | 5.21 | 11.88 |
| | | 1600 | 7.76 | 2.20 | 7.50 | 3.92 | 7.18 | 5.26 | 6.86 | 6.66 | 6.61 | 8.06 | 6.37 | 9.52 | 6.16 | 10.98 | 5.97 | 12.44 |
| | | 1700 | 8.26 | 2.32 | 8.00 | 4.15 | 7.67 | 5.53 | 7.36 | 7.04 | 7.08 | 8.53 | 6.83 | 10.04 | 6.61 | 11.54 | 6.42 | 13.04 |
| | | 1800 | 8.83 | 2.42 | 8.52 | 4.37 | 8.17 | 5.87 | 7.86 | 7.49 | 7.59 | 8.98 | 7.33 | 10.59 | 7.12 | 12.21 | 6.92 | 13.82 |
| | | 1900 | 9.32 | 2.52 | 8.97 | 4.83 | 8.64 | 6.20 | 8.33 | 7.82 | 8.06 | 9.43 | 7.81 | 11.15 | 7.59 | 12.63 | 7.39 | 14.14 |
| 100 | 4(100) | 800 | 5.08 | 1.37 | 4.51 | 2.65 | 4.00 | 3.80 | 3.54 | 4.88 | 3.14 | 5.68 | 2.79 | 6.78 | 2.48 | 7.62 | 2.20 | 8.43 |
| | | 900 | 5.91 | 1.55 | 5.33 | 3.00 | 4.82 | 4.31 | 4.36 | 5.52 | 3.96 | 6.55 | 3.60 | 7.67 | 3.29 | 8.63 | 3.01 | 9.54 |
| | | 1000 | 7.08 | 1.83 | 6.47 | 3.52 | 5.93 | 5.06 | 5.44 | 6.48 | 5.02 | 7.79 | 4.64 | 9.00 | 4.31 | 10.13 | 4.02 | 11.18 |
| | | 1100 | 7.92 | 2.01 | 7.31 | 3.87 | 6.77 | 5.56 | 6.28 | 7.12 | 5.86 | 8.67 | 5.48 | 9.90 | 5.15 | 11.14 | 4.75 | 12.29 |
| | | 1200 | 8.76 | 2.19 | 8.15 | 4.22 | 7.61 | 6.07 | 7.12 | 7.77 | 6.70 | 9.55 | 6.32 | 10.80 | 5.99 | 12.15 | 5.70 | 13.41 |
| | | 1300 | 9.60 | 2.38 | 8.99 | 4.57 | 8.45 | 6.58 | 7.96 | 8.42 | 7.54 | 10.43 | 7.16 | 11.70 | 6.83 | 13.17 | 6.54 | 14.53 |
| | | 1400 | 10.44 | 2.56 | 9.83 | 4.92 | 9.29 | 7.08 | 8.86 | 9.07 | 8.38 | 11.31 | 8.00 | 12.60 | 7.67 | 14.18 | 7.38 | 15.69 |
| | | 1500 | 11.34 | 2.67 | 10.75 | 4.93 | 10.23 | 7.59 | 9.75 | 9.72 | 9.25 | 12.22 | 8.88 | 13.50 | 8.56 | 15.19 | 8.28 | 16.75 |
| | | 1600 | 12.17 | 2.89 | 11.58 | 5.45 | 11.05 | 8.09 | 10.58 | 10.37 | 10.17 | 12.73 | 9.80 | 14.40 | 9.48 | 16.20 | 9.20 | 17.87 |
| | | 1700 | 13.00 | 3.12 | 12.41 | 5.97 | 11.88 | 8.60 | 11.40 | 11.01 | 11.00 | 13.23 | 10.63 | 15.30 | 10.31 | 17.21 | 10.03 | 18.99 |
| | | 1800 | 13.83 | 3.34 | 13.42 | 6.50 | 13.01 | 9.10 | 12.53 | 11.66 | 12.12 | 13.87 | 11.70 | 16.20 | 11.37 | 18.22 | 11.06 | 20.11 |
| | | 1900 | 14.65 | 3.57 | 14.15 | 7.03 | 13.75 | 9.61 | 13.32 | 12.31 | 12.91 | 14.25 | 12.54 | 17.10 | 12.22 | 19.23 | 11.94 | 21.23 |

Note: 1. The values are expressed as required power including drive power for the transmission unit.
2. The rated range for the air volume at specified ± 5%.
3. When the design pressure is 8000 mmAq and above, shall be applied RW type (water-cooling design)

RL

# RL PERFORMANCE TABLE (FOR PRESSURE OPERATION)

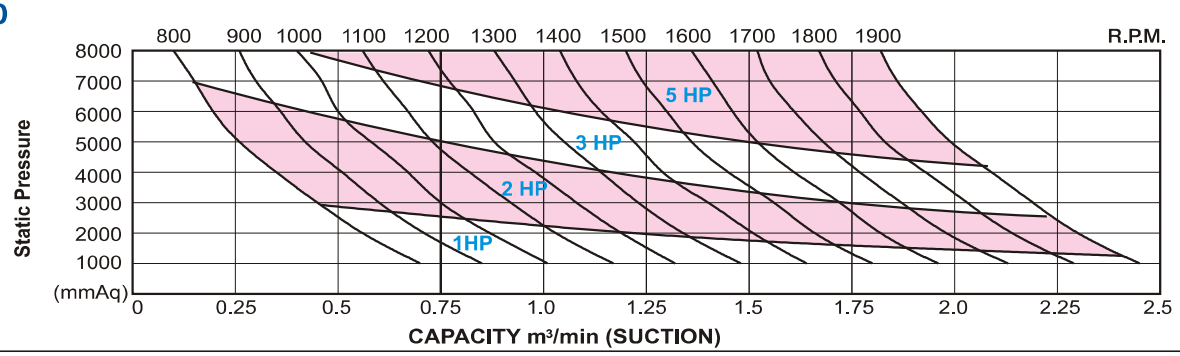
■ RLC - Complete Set ■ RL - Blower Body only

| TYPE | Bore | Revolution | 1000mmAq | | 2000mmAq | | 3000mmAq | | 4000mmAq | | 5000mmAq | | 6000mmAq | | 7000mmAq | | 8000mmAq | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 9.8kpa | | 19.6kpa | | 29.4kpa | | 39.4kpa | | 49.0kpa | | 58.8kpa | | 68kpa | | 78.4kpa | |
| | inch(mm) | R.P.M. | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw | m³/min | kw |
| 125 | 5(125) | 700 | 9.94 | 3.31 | 9.06 | 5.36 | 8.38 | 7.31 | 7.90 | 9.35 | 7.40 | 11.31 | 6.91 | 13.35 | 6.55 | 15.31 | 6.19 | 17.27 |
| | | 800 | 11.52 | 3.78 | 10.57 | 6.07 | 9.84 | 8.38 | 9.32 | 10.68 | 8.79 | 12.98 | 8.27 | 15.29 | 7.86 | 17.59 | 7.52 | 19.89 |
| | | 900 | 13.14 | 4.20 | 12.29 | 6.73 | 11.56 | 9.36 | 10.93 | 11.98 | 10.40 | 14.61 | 9.99 | 17.13 | 9.56 | 19.65 | 9.19 | 22.18 |
| | | 1000 | 15.05 | 4.55 | 14.02 | 7.45 | 13.34 | 10.34 | 12.74 | 13.24 | 12.20 | 16.03 | 11.72 | 18.92 | 11.79 | 21.92 | 10.67 | 24.82 |
| | | 1100 | 16.72 | 5.10 | 15.80 | 8.25 | 15.10 | 11.29 | 14.42 | 14.55 | 13.90 | 17.81 | 13.38 | 21.07 | 12.92 | 24.98 | 12.40 | 28.90 |
| | | 1200 | 18.45 | 5.32 | 17.51 | 8.86 | 16.79 | 12.31 | 16.23 | 15.74 | 15.74 | 19.18 | 15.26 | 22.93 | 14.80 | 26.69 | 14.30 | 30.44 |
| | | 1300 | 20.11 | 5.77 | 19.15 | 9.62 | 18.40 | 13.27 | 17.87 | 17.32 | 17.32 | 20.75 | 16.79 | 24.81 | 16.37 | 28.88 | 15.86 | 32.19 |
| | | 1400 | 21.77 | 6.23 | 20.78 | 10.28 | 20.01 | 14.22 | 19.47 | 18.27 | 18.93 | 22.97 | 18.38 | 27.35 | 17.94 | 30.63 | 17.42 | 33.91 |
| | | 1500 | 23.49 | 6.65 | 22.50 | 10.97 | 21.73 | 15.30 | 21.05 | 19.50 | 20.51 | 24.38 | 20.06 | 28.82 | 19.61 | 35.46 | 19.08 | 37.69 |
| | | 1600 | 25.27 | 7.06 | 24.33 | 11.59 | 23.70 | 16.21 | 23.06 | 20.74 | 22.54 | 26.33 | 21.98 | 30.54 | 21.59 | 34.75 | 21.24 | 38.97 |
| | | 1700 | 27.01 | 7.38 | 26.05 | 12.35 | 25.32 | 17.19 | 24.79 | 23.21 | 24.27 | 27.43 | 23.73 | 32.71 | 23.31 | 36.92 | 22.93 | 41.43 |
| | | 1800 | 28.74 | 8.17 | 27.79 | 13.11 | 27.05 | 18.18 | 26.52 | 24.30 | 26.00 | 29.59 | 25.46 | 34.87 | 25.08 | 39.04 | 24.71 | 43.20 |
| 150 | 6(150) | 700 | 13.09 | 3.99 | 11.97 | 6.65 | 11.25 | 9.21 | 10.54 | 11.77 | 9.93 | 14.42 | 9.41 | 16.98 | 8.98 | 19.54 | 8.66 | 22.09 |
| | | 800 | 15.06 | 4.62 | 13.96 | 7.59 | 13.19 | 10.44 | 12.41 | 13.41 | 11.76 | 16.38 | 11.21 | 19.45 | 10.71 | 22.53 | 10.36 | 25.60 |
| | | 900 | 17.32 | 4.96 | 16.22 | 8.38 | 15.34 | 11.81 | 14.56 | 15.12 | 14.01 | 18.43 | 13.35 | 22.06 | 12.89 | 25.70 | 12.35 | 29.34 |
| | | 1000 | 19.54 | 5.65 | 18.46 | 9.33 | 17.59 | 13.03 | 16.93 | 16.72 | 16.29 | 20.41 | 15.64 | 23.89 | 15.09 | 27.36 | 14.80 | 30.95 |
| | | 1100 | 21.89 | 6.15 | 20.75 | 10.26 | 19.84 | 14.36 | 19.04 | 18.47 | 18.47 | 22.57 | 17.78 | 27.38 | 17.31 | 32.20 | 16.94 | 37.00 |
| | | 1200 | 23.97 | 6.67 | 22.88 | 11.05 | 21.99 | 15.53 | 21.34 | 19.92 | 20.69 | 25.17 | 20.14 | 29.55 | 19.69 | 33.93 | 19.33 | 38.30 |
| | | 1300 | 26.16 | 7.19 | 24.93 | 12.58 | 24.14 | 16.73 | 23.35 | 21.56 | 22.79 | 26.95 | 22.12 | 31.44 | 21.67 | 35.93 | 21.33 | 40.42 |
| | | 1400 | 28.24 | 7.69 | 27.09 | 12.86 | 26.18 | 18.03 | 25.49 | 24.07 | 24.80 | 28.71 | 24.22 | 34.45 | 23.79 | 40.19 | 23.41 | 46.94 |
| | | 1500 | 30.48 | 8.26 | 29.32 | 13.72 | 28.39 | 19.31 | 27.57 | 25.59 | 26.99 | 31.41 | 26.41 | 37.22 | 25.93 | 43.04 | 25.57 | 48.86 |
| | | 1600 | 32.83 | 8.74 | 31.73 | 14.59 | 30.85 | 20.56 | 30.07 | 27.64 | 29.52 | 33.16 | 28.96 | 37.73 | 28.52 | 43.26 | 28.17 | 48.79 |
| | | 1700 | 34.99 | 9.19 | 33.88 | 15.50 | 33.00 | 21.85 | 32.34 | 28.79 | 31.67 | 35.44 | 31.11 | 40.97 | 30.61 | 46.41 | 30.26 | 52.05 |
| | | 1800 | 37.26 | 9.20 | 36.16 | 16.41 | 35.27 | 23.29 | 34.49 | 29.95 | 33.94 | 36.60 | 33.26 | 43.26 | 32.75 | 49.91 | 32.39 | 56.57 |
| 200 | 8(200) | 600 | 23.52 | 6.77 | 21.87 | 11.52 | 20.52 | 16.26 | 19.45 | 21.29 | 18.58 | 25.16 | 17.61 | 30.97 | 16.84 | 35.81 | 16.16 | 40.65 |
| | | 700 | 27.90 | 7.90 | 26.20 | 13.40 | 24.90 | 18.90 | 23.80 | 24.00 | 22.80 | 30.00 | 21.90 | 37.00 | 23.00 | 44.00 | 22.30 | 47.00 |
| | | 800 | 32.67 | 9.06 | 31.04 | 15.33 | 29.69 | 22.17 | 28.63 | 27.95 | 27.66 | 35.66 | 26.89 | 41.45 | 26.12 | 47.23 | 25.45 | 53.01 |
| | | 900 | 37.26 | 10.06 | 35.61 | 17.13 | 34.26 | 24.19 | 33.19 | 30.97 | 32.32 | 38.71 | 31.45 | 45.48 | 30.68 | 52.26 | 30.00 | 59.03 |
| | | 1000 | 41.90 | 11.05 | 40.29 | 19.05 | 38.95 | 26.67 | 37.90 | 35.24 | 37.05 | 42.86 | 36.19 | 50.48 | 35.43 | 58.10 | 34.76 | 65.71 |
| | | 1100 | 46.70 | 12.20 | 45.00 | 21.00 | 43.70 | 30.00 | 42.60 | 39.00 | 41.60 | 48.00 | 40.70 | 56.00 | 39.90 | 65.00 | 39.20 | 74.00 |
| | | 1200 | 50.98 | 13.25 | 49.34 | 23.04 | 48.10 | 31.68 | 47.04 | 42.24 | 46.08 | 50.88 | 45.22 | 60.48 | 44.45 | 69.12 | 43.78 | 79.68 |
| | | 1300 | 55.45 | 14.28 | 53.77 | 24.62 | 52.49 | 34.47 | 51.41 | 45.30 | 50.42 | 55.15 | 49.54 | 65.00 | 48.75 | 75.83 | 48.06 | 86.67 |
| | | 1400 | 59.93 | 15.31 | 58.22 | 26.19 | 56.91 | 38.27 | 55.80 | 48.35 | 54.79 | 59.42 | 53.99 | 70.50 | 53.08 | 81.58 | 52.05 | 92.66 |
| 250 | 10(250) | 600 | 31.94 | 8.82 | 30.00 | 15.15 | 28.55 | 21.78 | 27.29 | 29.05 | 26.13 | 34.24 | 25.06 | 41.50 | 24.37 | 46.68 | 23.71 | 51.87 |
| | | 700 | 37.80 | 10.29 | 35.90 | 17.69 | 34.30 | 25.73 | 33.00 | 32.16 | 32.00 | 40.74 | 30.90 | 48.24 | 29.90 | 54.67 | 29.20 | 61.10 |
| | | 800 | 44.24 | 11.57 | 42.22 | 20.14 | 40.67 | 28.92 | 39.52 | 37.20 | 38.46 | 46.50 | 37.40 | 53.73 | 36.43 | 62.00 | 35.93 | 70.27 |
| | | 900 | 50.32 | 12.97 | 48.39 | 22.82 | 46.94 | 31.12 | 45.77 | 41.50 | 44.52 | 50.84 | 43.45 | 60.17 | 42.77 | 69.50 | 42.10 | 78.83 |
| | | 1000 | 56.48 | 14.29 | 54.67 | 24.50 | 53.05 | 34.71 | 51.90 | 45.94 | 50.76 | 56.19 | 49.81 | 66.55 | 48.95 | 77.59 | 48.19 | 88.63 |
| | | 1100 | 63.00 | 15.75 | 61.00 | 27.87 | 59.40 | 39.66 | 58.10 | 51.46 | 57.00 | 62.18 | 56.00 | 73.97 | 55.00 | 85.76 | 54.20 | 97.55 |
| | | 1200 | 68.64 | 17.09 | 66.82 | 29.85 | 65.28 | 43.22 | 64.03 | 56.46 | 62.88 | 66.89 | 61.92 | 80.28 | 61.06 | 92.62 | 60.19 | 104.93 |
| | | 1300 | 74.65 | 18.48 | 72.78 | 31.67 | 71.20 | 46.46 | 69.92 | 59.09 | 68.74 | 72.85 | 67.76 | 86.57 | 66.87 | 99.94 | 66.08 | 113.36 |
| | | 1400 | 80.68 | 19.86 | 78.56 | 34.55 | 77.05 | 49.66 | 75.90 | 63.71 | 74.63 | 78.82 | 73.63 | 93.93 | 72.52 | 109.08 | 71.71 | 124.19 |
| 300 | 12(300) | 600 | 71.2 | 18.1 | 68.0 | 33.8 | 65.3 | 48.1 | 62.6 | 61.4 | 60.5 | 73.6 | 58.5 | 84.9 | 56.8 | 95.5 | 55.3 | 105.2 |
| | | 700 | 85.5 | 20.8 | 81.4 | 39.2 | 78.7 | 55.9 | 76.0 | 71.4 | 73.9 | 85.6 | 71.9 | 98.8 | 70.2 | 111.1 | 68.7 | 122.5 |
| | | 800 | 98.5 | 23.7 | 94.8 | 44.7 | 92.1 | 63.8 | 89.4 | 81.4 | 87.3 | 97.7 | 85.3 | 112.7 | 83.6 | 126.8 | 82.1 | 139.8 |
| | | 900 | 111.5 | 26.6 | 108.2 | 50.1 | 105.6 | 71.6 | 102.9 | 91.4 | 100.8 | 109.7 | 98.8 | 126.7 | 97.1 | 142.4 | 95.6 | 157.1 |
| | | 1000 | 124.5 | 29.6 | 121.6 | 57.0 | 118.9 | 80.9 | 116.2 | 102.9 | 114.1 | 123.3 | 112.1 | 142.1 | 110.4 | 159.6 | 108.9 | 176.0 |
| | | 1100 | 138.1 | 34.5 | 135.1 | 63.9 | 132.4 | 90.2 | 129.7 | 114.4 | 127.6 | 136.8 | 125.6 | 157.6 | 123.9 | 176.9 | 122.4 | 194.8 |
| | | 1200 | 151.7 | 39.5 | 148.5 | 70.8 | 145.8 | 99.5 | 143.1 | 125.9 | 141.0 | 150.4 | 139.0 | 171.9 | 137.3 | 194.1 | 135.8 | 213.7 |
| | | 1300 | 165.3 | 44.5 | 162.1 | 83.6 | 159.4 | 108.8 | 156.7 | 137.4 | 154.6 | 164.0 | 152.6 | 188.4 | 150.9 | 211.3 | 149.4 | 232.6 |
| | | 1400 | 178.0 | 47.9 | 174.6 | 90.0 | 171.7 | 117.2 | 168.8 | 148.0 | 166.5 | 176.6 | 164.3 | 202.9 | 162.5 | 227.6 | 160.9 | 250.5 |

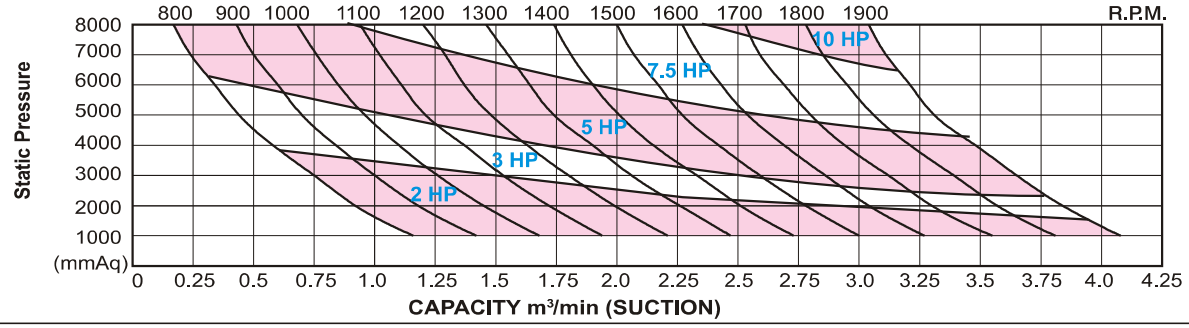
# RL PERFORMANCE CURVE (FOR PRESSURE OPERATION)

■ RLC - Complete Set ■ RL - Blower Body only

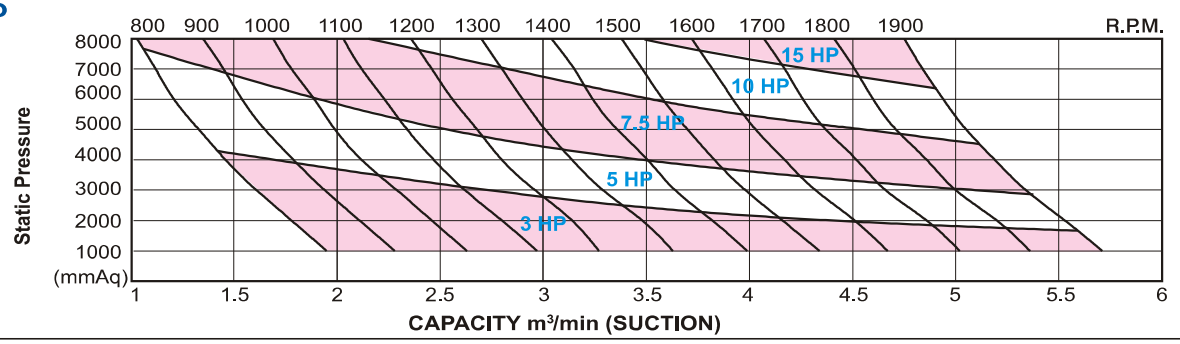
## 40



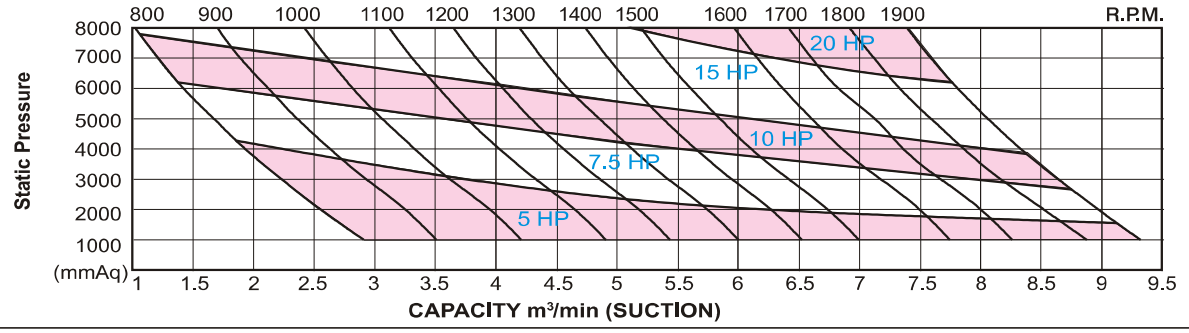

## 50



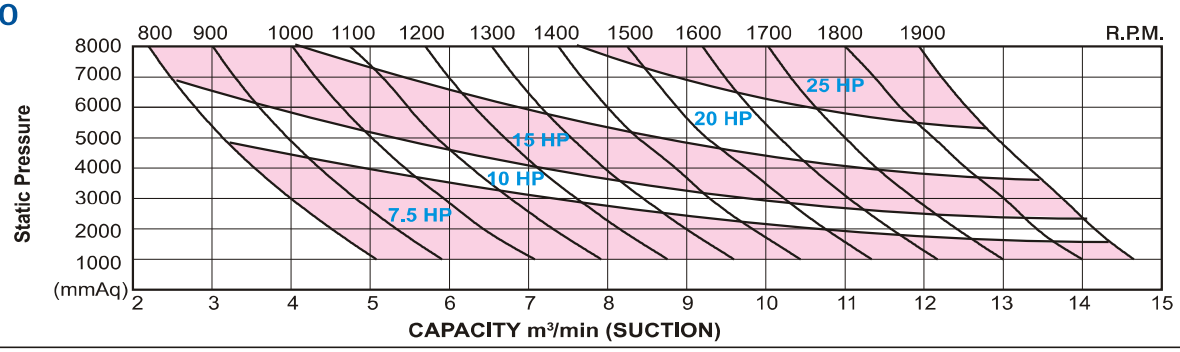

## 65




## 80




## 100

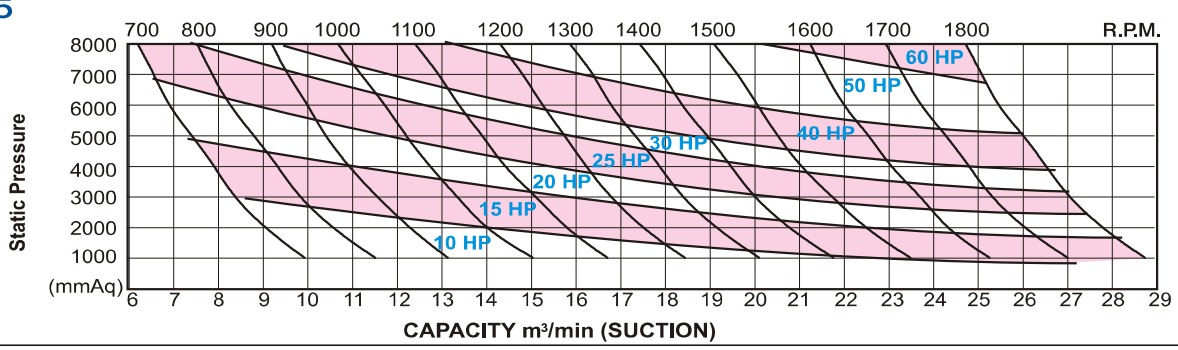



RL

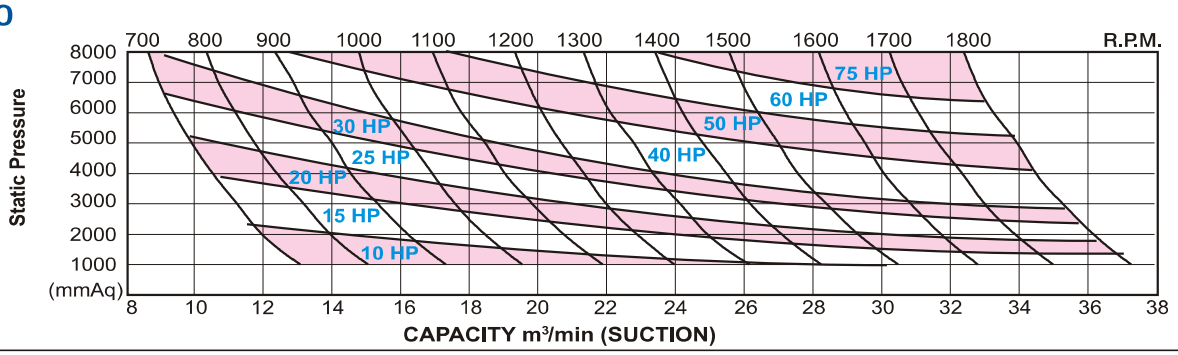
# RL PERFORMANCE CURVE (FOR PRESSURE OPERATION)

■ RLC - Complete Set ■ RL - Blower Body only

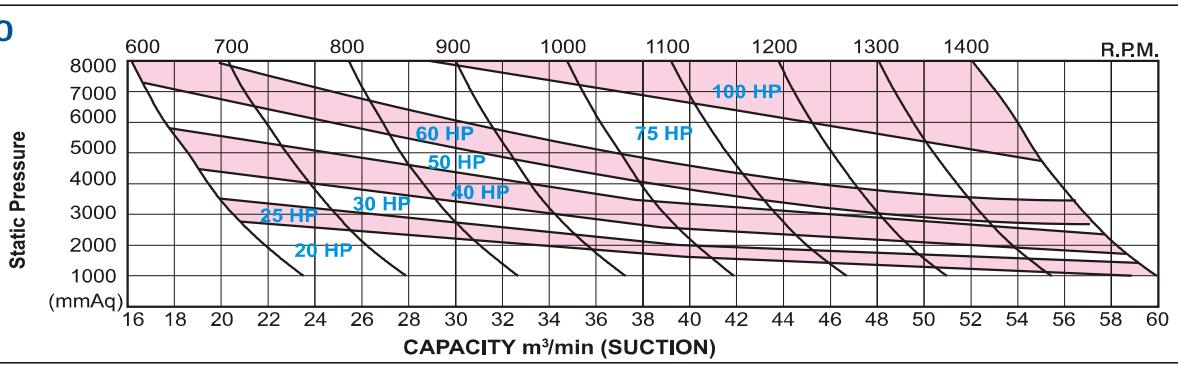
## 125



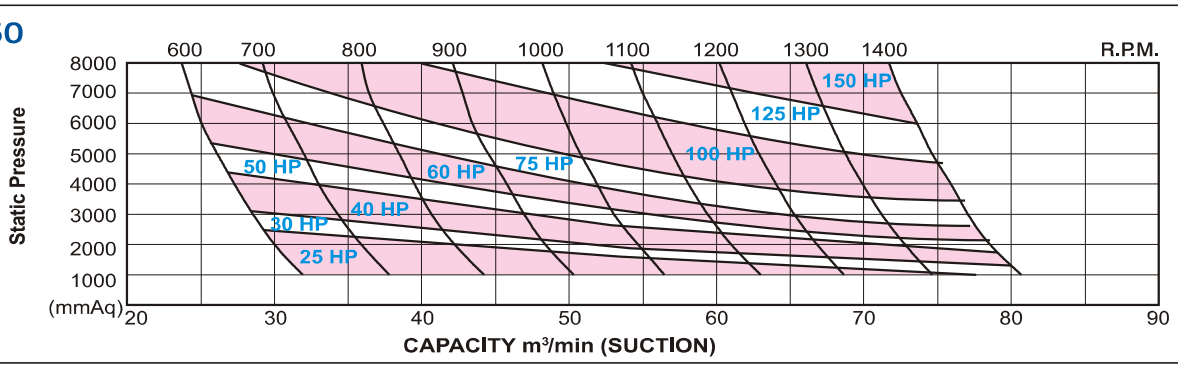

## 150



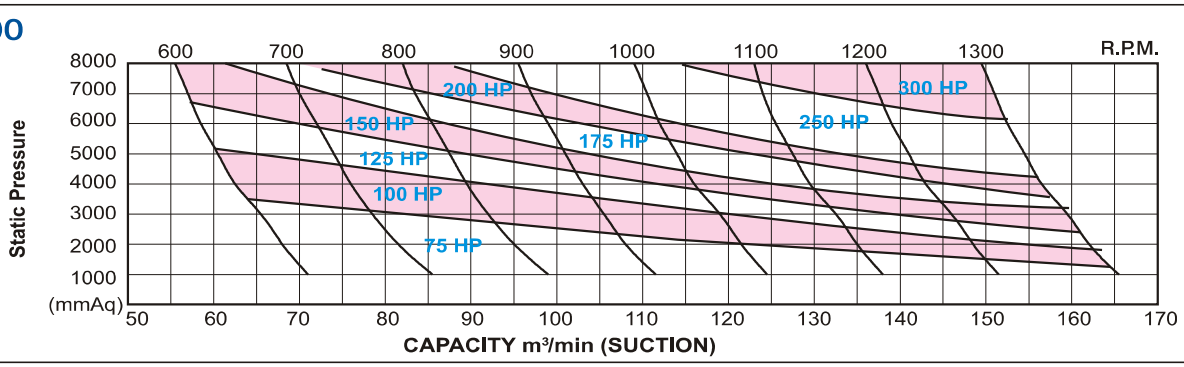

## 200




## 250




## 300



# HOW TO USE PERFORMANCE TABLES

The performance tables give the model number, bore, r.p.m. , static pressure, air volume and required power of the motor.

1. The air volume in the performance table are the volume of air intake under standard conditions. Standard conditions are temperature 20 °C, absolute pressure 1 atm (10333mmAq) and relative humidity 65%.
2. Air volume is generally shown in terms of following three modes. Values can be converted between them.
   1 ) Air volume at **suction condition**
   Shown in terms of suction pressure, temperature and humidity.
   2 ) Air volume at **standard condition**
   3 ) Air volume at **base condition**
   When the temperature is 0 °C, absolute pressure 1 atm and when the air is dry.
3. The air volume conversion formula is as follows :
   $Q_2 = Q_1 \times P_1 / P_2 \times T_2 / T_1$
   $Q_1$ : air volume($m^3$/min)at absolute pressure of $P_1$ (mmAq) and absolute temperature of $T_1$ (K).
   $Q_2$ : air volume($m^3$/min)at absolute pressure of $P_2$ (mmAq) and absolute temperature of $T_2$ (K).
4. According to the air volume and static pressure as calculated above, the model number, bore, r.p.m. and required power can be found in the performance table.
5. The choice is overlapped depending upon the type of blower. For reference, however, selection should be lower number blowers for the economy and higher number for the sound level.
6. The values for power displayed in the performance table are expressed as required power including drive power for the transmission unit, so there is no need to create a margin when selecting a motor.
7. power (output) can be calculated using the following formula
   $kW = \sqrt{3} \times V \times A \times \cos\theta \times Eff. \times 10^{-3}$
   V : voltage A : current cos. θ : motor power factor
   Eff. : motor efficiency

| PRESSURE | atm | kpa | mbar | ℓbf/m²(psi) | kgf/cm² | in Hg | ft Aq | mmHg(Torr) | mmAq |
|---|---|---|---|---|---|---|---|---|---|
| 1 atm | 1 | 101.325 | 1013.25 | 14.696 | 1.0333 | 29.921 | 33.914 | 760 | 10333 |
| 1 kpa | 0.0099 | 1 | 10 | 0.145 | 0.0102 | 0.295 | 0.335 | 7.5 | 102 |
| 1 mbar | 0.00099 | 0.1 | 1 | 0.0145 | 0.00102 | 0.0295 | 0.0335 | 0.75 | 10.198 |
| 1 ℓbf/m²(psi) | 0.069 | 6.894 | 68.965 | 1 | 0.0703 | 2.036 | 2.308 | 51.71 | 703 |
| 1 kgf/cm² | 0.968 | 98.062 | 980.392 | 14.228 | 1 | 28.96 | 32.82 | 735.53 | 10000 |
| 1 inHg | 0.0334 | 3.3863 | 33.898 | 0.491 | 0.0345 | 1 | 1.133 | 25.4 | 345.3 |
| 1 ftAq | 0.0295 | 2.99 | 29.851 | 0.434 | 0.0305 | 0.882 | 1 | 22.42 | 304.8 |
| 1 mmHg(Torr) | 0.00132 | 0.1333 | 1.333 | 0.019 | 0.0014 | 0.04 | 0.045 | 1 | 13.6 |
| 1 mmAq | 0.000097 | 0.0098 | 0.09803 | 0.0014 | 0.0001 | 0.003 | 0.003 | 0.074 | 1 |

| CAPACITY | m³/min | ℓ/min | cm³/sec | in³/sec | ft³/min(CFM) | U.S. GPM |
|---|---|---|---|---|---|---|
| 1m³/min | 1 | 1000 | 16667 | 1016 | 35.288 | 264.172 |
| 1ℓ/min | 0.001 | 1 | 16.67 | 1.02 | 0.0353 | 0.2641 |
| 1cm³/sec | 0.00006 | 0.06 | 1 | 0.061 | 0.002 | 0.0158 |
| 1in³/sec | 0.00098 | 0.983 | 16.39 | 1 | 0.035 | 0.2589 |
| 1ft³/min(CFM) | 0.028 | 28.32 | 471.957 | 28.8 | 1 | 7.4836 |
| 1U.S. GPM | 0.003785 | 3.785 | 63.0915 | 3.8460 | 0.13360 | 1 |

**Pressure Conversion Formula**

1 kpa=1000 pa=1000 N/M²

1 bar=1000 mbar=100 kpa

1 mbar=100 N/M²

# NOISE LEVEL - RL 、RW TYPE

perotac MADE IN TAIWAN

**1. Measure method (1.0m On Machine Side)**

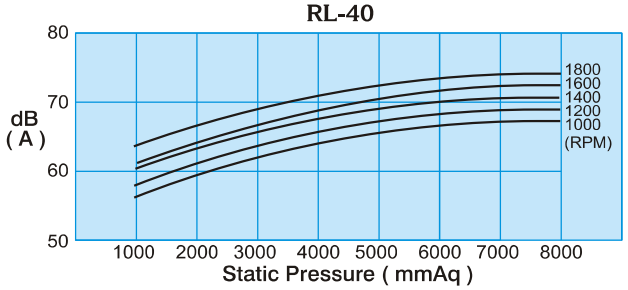
For your information, the following noise reading are the average value measured at a distance 1m from the PEROTAC blowers under standard condition.

**2. Noise prevention method**

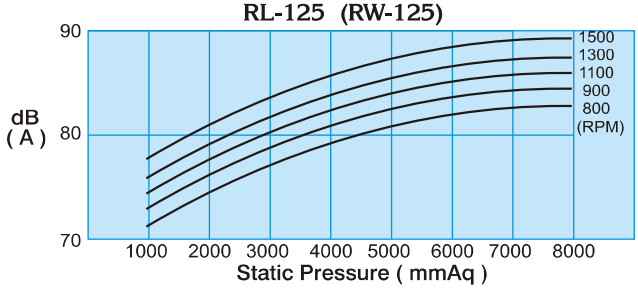
The noise of PEROTAC blower has direct relationship with the factors such as blower size, discharge pressure and R.P.M. To get even greater effectiveness in noise reduction, please refer to some methods as follow :

1. Establish the silencer to inlet and outlet to prevent the noise of air flow.
2. Decrease the revolution of blower to reduce the noise of it.
3. Use the expansion joint or the rubber vibration isolator to prevent the surging and vibration.

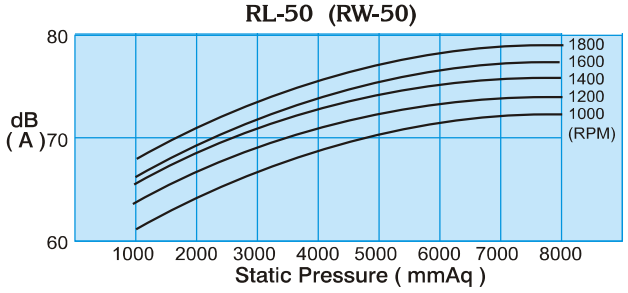
RL-40



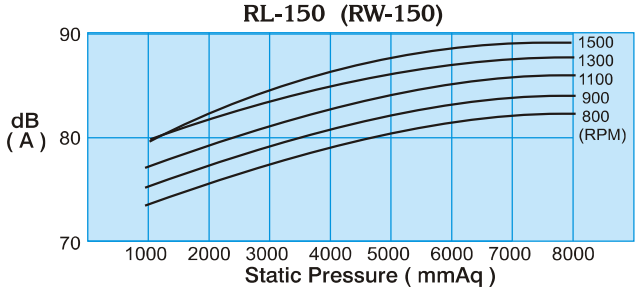

RL-125 (RW-125)



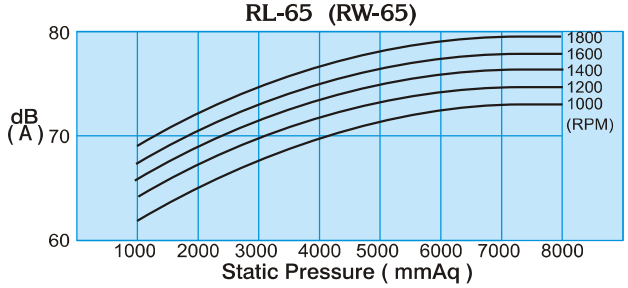

RL-50 (RW-50)



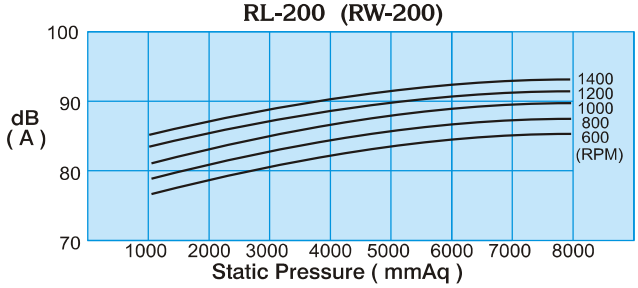

RL-150 (RW-150)



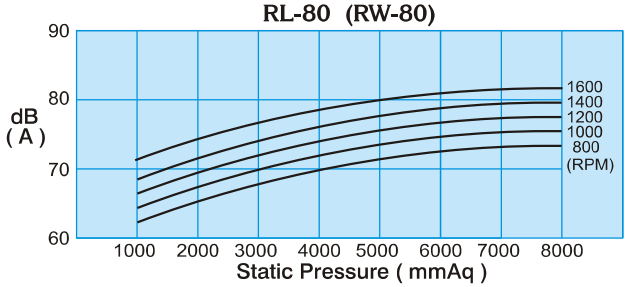

RL-65 (RW-65)



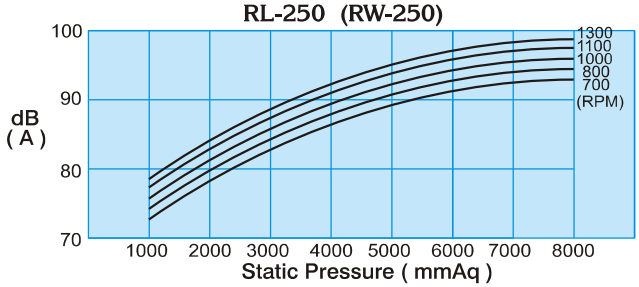

RL-200 (RW-200)



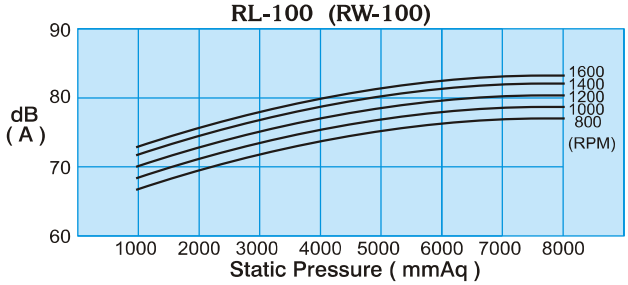

RL-80 (RW-80)



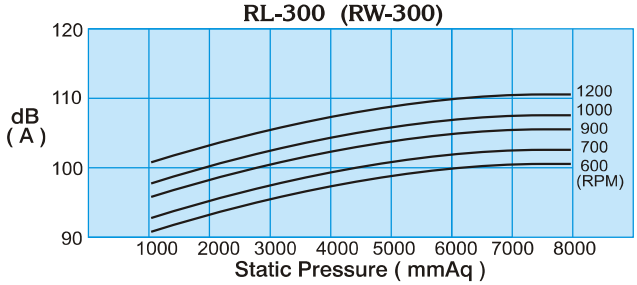

RL-250 (RW-250)



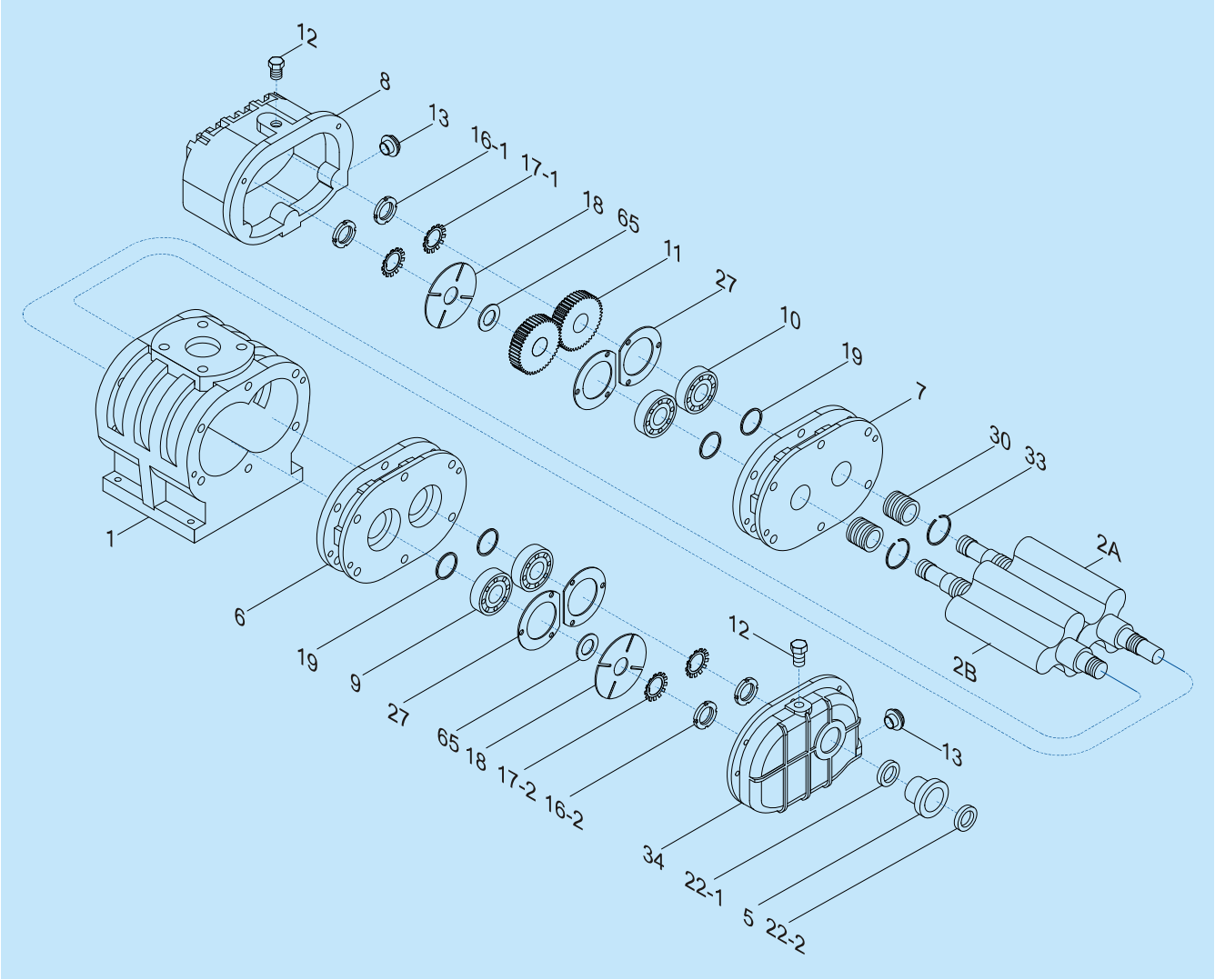

RL-100 (RW-100)




RL-300 (RW-300)



# RL STRUCTURAL DRAWING




## PARTS LIST

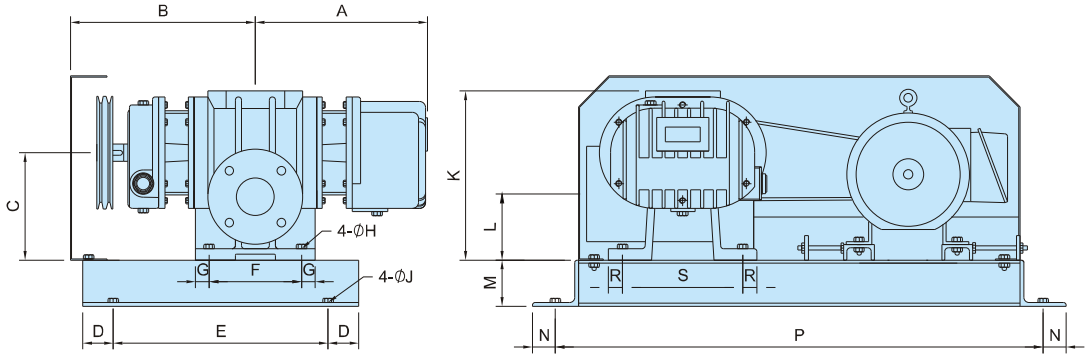
| NO. | NAME | MATERIAL | Q'ty | FITTING MODEL |
|---|---|---|---|---|
| 1 | CASING | FC25 | 1 | |
| 2A | ROTOR+DRIVE SHAFT | FC25+SCM440 | 1 | |
| 2B | ROTOR+DRIVEN SHAFT | FC25+SCM440 | 1 | |
| 5 | SEAL COVER | FC25 | 4 | RL-80~300 |
| 6 | FRONT SIDE COVER | FC25 | 1 | |
| 7 | REAR SIDE COVER | FC25 | 1 | |
| 8 | GEAR BOX | FC25 | 1 | |
| 9 | FRONT BEARING | SUJ2 | 2 | |
| 10 | REAR BEARING | SUJ2 | 2 | |
| 11 | TIMING GEAR | SNCM220 | 2 | |
| 12 | LUBRICATION PLUG | S45C | 2 | |
| 13 | OIL GAUGE | GLASS | 2 | |
| | | | | |

| NO. | NAME | MATERIAL | Q'ty | FITTING MODEL |
|---|---|---|---|---|
| 16-1 | LOCK NUT | S45C | 2 | |
| 16-2 | LOCK NUT | S45C | 2 | |
| 17-1 | SPRING WASHER | SS41 | 2 | |
| 17-2 | SPRING WASHER | SS41 | 2 | |
| 18 | OIL SPLASH | SS41 | 2 | |
| 19 | V SEAL | VITON | 4 | |
| 22-1 | OIL SEAL | VITON | 1 | RL- 40~300 |
| 22-2 | OIL SEAL | VITON | 1 | RL-125~300 |
| 27 | BEARING COVER | SS41 | 4 | |
| 30 | BEARING HOLDER | SCM440 | 2 | RL-150,RL-300 |
| 33 | PISTON RING | FC25 | 2 | (OPTIONAL) |
| 34 | FRONT COVER | FC25 | 1 | |
| 65 | WASHER | SS41 | 2 | |

# OUTLINE DIMENSIONS

## RL TYPE

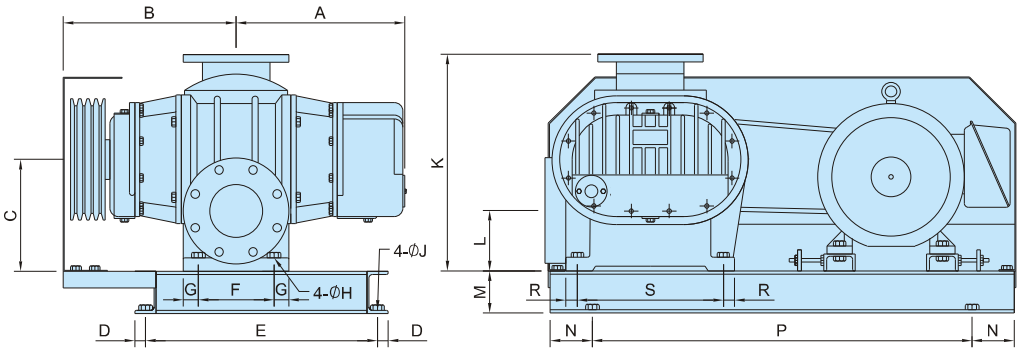
### ◆ DIMENSIONS OF RL-40,50, 65




UNIT : mm

| TYPE | A | B | C | D | E | F | G | H | J | K | L | M | N | P | R | S | Wt.(kg) | RLC(kg) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RL-40 | 238 | 260 | 180 | 20 | 340 | 80 | 15 | 14 | 14 | 290 | 104 | 46 | 41.5 | 517 | 23 | 196 | 65 | 103 |
| RL-50 | 278 | 302 | 175 | 50 | 350 | 150 | 22.5 | 14 | 14 | 276 | 103.5 | 75 | 37.5 | 795 | 22 | 196 | 70 | 136 |
| RL-65 | 308 | 323 | 173 | 50 | 350 | 195 | 30.5 | 14 | 14 | 276 | 104 | 75 | 37.5 | 795 | 25 | 200 | 75 | 155 |

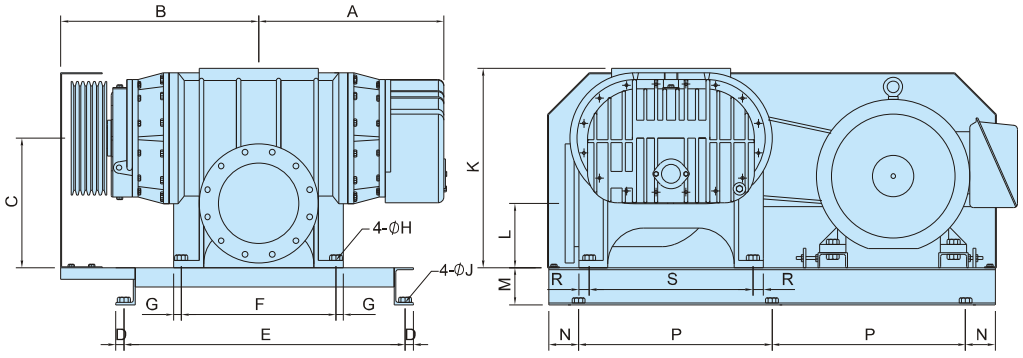
### ◆ DIMENSIONS OF RL-80, 100, 125, 150




UNIT : mm

| TYPE | A | B | C | D | E | F | G | H | J | K | L | M | N | P | R | S | Wt.(kg) | RLC(kg) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RL-80 | 320 | 406 | 213 | 50 | 450 | 195 | 22.5 | 16 | 14 | 341 | 106.5 | 75 | 37.5 | 945 | 20 | 200 | 143 | 250 |
| RL-100 | 320 | 467 | 225 | 25 | 550 | 282 | 20.5 | 16 | 19 | 354 | 114.5 | 100 | 100 | 760 | 30 | 232 | 166 | 330 |
| RL-125 | 455 | 500 | 265 | 25 | 550 | 310 | 30 | 19 | 19 | 498 | 157.5 | 100 | 100 | 900 | 22.5 | 355 | 296 | 485 |
| RL-150 | 510 | 566 | 295 | 25 | 550 | 428 | 26 | 19 | 19 | 457 | 163 | 100 | 100 | 900 | 22.5 | 355 | 352 | 525 |

### ◆ DIMENSIONS OF RL-200, 250, 300



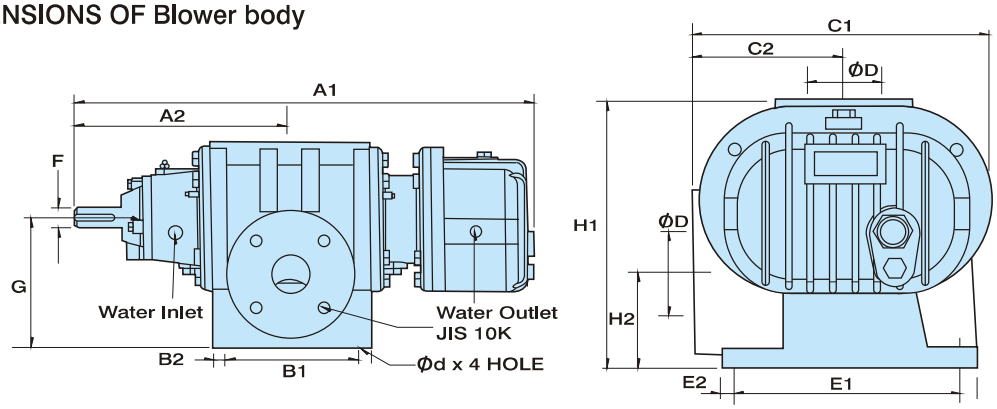

UNIT : mm

| TYPE | A | B | C | D | E | F | G | H | J | K | L | M | N | P | R | S | Wt.(kg) | RLC(kg) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RL-200 | 547 | 620 | 405 | 32.5 | 935 | 400 | 40 | 23 | 24 | 705 | 194 | 125 | 100 | 650 | 33 | 550 | 880 | 1155 |
| RL-250 | 622 | 680 | 435 | 32.5 | 935 | 520 | 25 | 23 | 24 | 675 | 216 | 125 | 100 | 650 | 35 | 550 | 1050 | 1500 |
| RL-300 | 685 | 828 | 550 | 35 | 1035 | 510 | 30 | 23 | 24 | 1010 | 300 | 250 | 235 | 1000 | 35 | 780 | 2000 | 2800 |

# OUTLINE DIMENSIONS

## RW TYPE

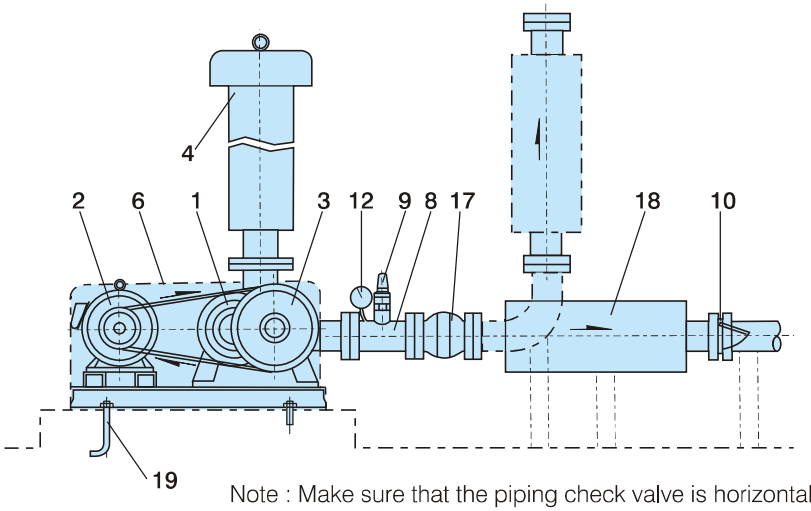
◆ DIMENSIONS OF Blower body




UNIT : mm

| TYPE | A1 | A2 | B1 | B2 | C1 | C2 | ϕD | ϕd | E1 | E2 | F | G | H1 | H2 | Wt. (kg) | RWC(kg) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RW-50 | 582 | 277 | 150 | 22.5 | 294 | 158 | 50 | 14 | 196 | 22 | 28 | 175 | 276 | 104 | 70 | 136 |
| RW-65 | 642 | 307 | 195 | 30.5 | 288 | 152 | 65 | 14 | 200 | 25 | 28 | 175 | 276 | 104 | 75 | 155 |
| RW-80 | 705 | 305 | 195 | 22.5 | 358 | 180 | 80 | 16 | 200 | 20 | 38 | 213 | 341 | 107 | 143 | 250 |
| RW-100 | 805 | 355 | 282 | 20.5 | 364 | 186 | 100 | 16 | 232 | 30 | 38 | 225 | 354 | 115 | 166 | 330 |
| RW-125 | 900 | 437 | 310 | 30 | 483 | 250 | 125 | 19 | 355 | 22.5 | 48 | 265 | 498 | 158 | 296 | 485 |
| RW-150 | 1010 | 492 | 428 | 26 | 483 | 250 | 150 | 19 | 355 | 22.5 | 48 | 295 | 457 | 163 | 352 | 525 |
| RW-200 | 1137 | 590 | 400 | 40 | 685 | 345 | 200 | 23 | 550 | 33 | 65 | 405 | 705 | 194 | 880 | 1155 |
| RW-250 | 1287 | 665 | 520 | 25 | 715 | 356 | 250 | 23 | 550 | 35 | 65 | 435 | 675 | 216 | 1050 | 1500 |
| RW-300 | 1330 | 645 | 510 | 30 | 1060 | 560 | 300 | 33 | 780 | 35 | 95 | 550 | 1010 | 300 | 2000 | 2800 |

# REFERENCE DRAWING FOR PIPING (EXAMPLE)




Note : Make sure that the piping check valve is horizontal.

| NO | ACCESSORIES |
|---|---|
| 1 | Blower body |
| 2 | Motor |
| 3 | Blower pulley |
| 4 | Suction silencer set |
| 6 | Belt cover |
| 8 | T-type reducer |
| 9 | Pressure safety valve |
| 10 | Check valve |
| 12 | Pressure gauge |
| 17 | Expansion joint |
| 18 | Discharge silencer |
| 19 | Anchor bolt |

An example of piping arrangement around the blower is shown. Make the most suitable installation from the viewpoint of space, piping schedule, vibration prevention, etc. It is effective to use rubber vibration isolator and to provide an expansion joint to the blower outlet for the prevention of vibration. A silencer may be used to either laterally or vertically.
Install the check valve on the horizontal piping. Use a suitable support to the piping to avoid applying a piping load to the blower nozzle.

# DIMENSIONS OF ACCESSORIES

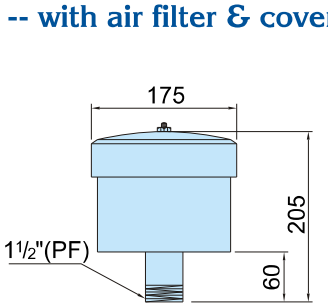
## BLOWER PULLEY (No.3)

| TYPE | BORE | DIMENSION(O.D.) |
|---|---|---|
| SPB170 x 1G | 40A | 170mm |
| SPB180 x 2G | 50A | 180mm |
| SPB180 x 2G | 65A | 180mm |
| SPB200 x 3G | 80A | 200mm |
| SPB200 x 4G | 100A | 200mm |

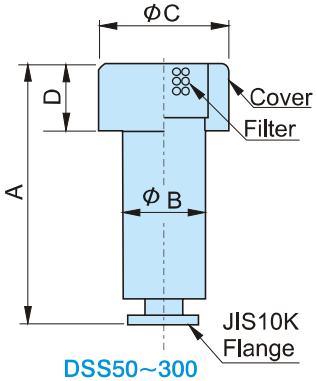
| TYPE | BORE | DIMENSION(O.D.) |
|---|---|---|
| SPB280 x 4G | 125A | 280mm |
| SPB280 x 4G | 150A | 280mm |
| SPB375 x 5G | 200A | 375mm |
| SPB375 x 6G | 250A | 375mm |
| SPC450 x 8G | 300A | 450mm |

## SUCTION & DISCHARGE SILENCER SET (No.4)

-- with air filter & cover




DSS40

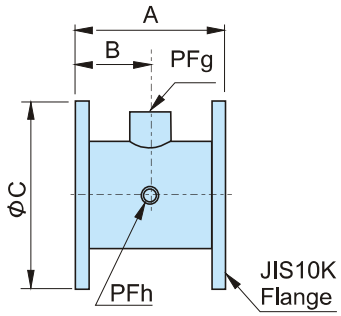



DSS50~300

UNIT : mm

| TYPE | BORE | A | B | C | D | WT.(kg) |
|---|---|---|---|---|---|---|
| DSS50 | 50A | 548 | 140 | 220 | 80 | 7.5 |
| DSS65 | 65A | 548 | 165 | 235 | 85 | 14 |
| DSS80 | 80A | 818 | 210 | 310 | 145 | 22 |
| DSS100 | 100A | 818 | 240 | 350 | 160 | 26 |
| DSS125 | 125A | 1310 | 280 | 400 | 15 | 30 |
| DSS150 | 150A | 1310 | 300 | 450 | 190 | 43 |
| DSS200 | 200A | 1500 | 400 | 500 | 210 | 85 |
| DSS250 | 250A | 1700 | 400 | 500 | 210 | 110 |
| DSS300 | 300A | 1824 | 500 | 650 | 350 | 130 |

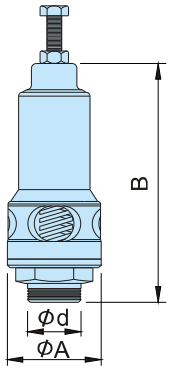
## T-TYPE REDUCER (No.8)




UNIT : mm

| TYPE | BORE | A | B | C | PFg | PFh | WT.(kg) |
|---|---|---|---|---|---|---|---|
| TTR50 | 40A.50A | 150 | 75 | 155 | PF 1 1/2" | PF 1/4" | 5 |
| TTR65 | 65A | 150 | 75 | 175 | PF 1 1/4" | PF 1/4" | 7 |
| TTR80 | 80A | 180 | 90 | 185 | PF 1 1/2" | PF 1/4" | 9 |
| TTR100 | 100A | 180 | 90 | 210 | PF 1 1/2" | PF 1/4" | 11 |
| TTR125 | 125A | 250 | 125 | 250 | PF 1 1/2" | PF 1/4" | 18 |
| TTR150 | 150A | 250 | 125 | 280 | PF 1 1/2" | PF 1/4" | 19 |
| TTR200 | 200A | 250 | 125 | 330 | PF 3" | PF 1/4" | 26 |
| TTR250 | 250A | 250 | 125 | 400 | PF 3" | PF 1/4" | 31 |
| TTR300 | 300A | 300 | 150 | 450 | PF 3" | PF 1/4" | 36 |

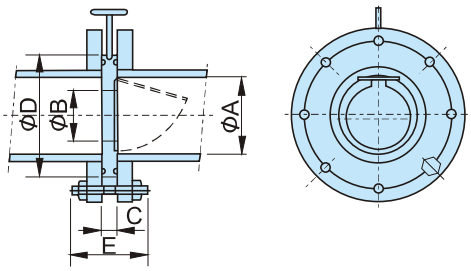
## PRESSURE SAFETY VALVE (No.9)




UNIT : mm

| TYPE | ΦA | B | Φd | Fitting Model | WT.(kg) |
|---|---|---|---|---|---|
| PSV 32 | 71.5 | 182 | PF1 1/4" | RLC/RWC -40,50,65,80 | 2 |
| PSV 50 | 85 | 197 | PF 2" | RLC/RWC -100,125,150 | 3 |
| PSV 80 | 115 | 221 | PF 3" | RLC/RWC -200,250,300 | 5.5 |

## CHECK VALVE (No.10)

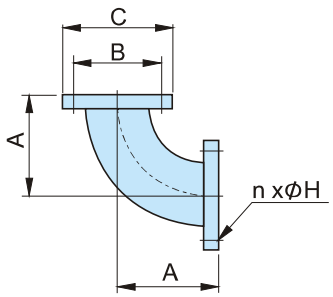



UNIT : mm

| TYPE | BORE | A | B | C | D | E | WT.(kg) |
|---|---|---|---|---|---|---|---|
| NRV50 | 40A.50A | 52.7 | 25 | 18 | 104 | 86 | 1 |
| NRV65 | 65A | 69.9 | 36 | 19 | 124 | 108 | 1.5 |
| NRV80 | 80A | 78.1 | 46 | 20 | 134 | 108 | 2 |
| NRV100 | 100A | 102.3 | 67 | 22 | 159 | 108 | 2.5 |
| NRV125 | 125A | 126.6 | 88 | 25 | 190 | 115 | 3.5 |
| NRV150 | 150A | 151.0 | 108 | 25 | 220 | 115 | 5 |
| NRV200 | 200A | 200.0 | 138 | 25 | 270 | 127 | 7 |
| NRV250 | 250A | 251.4 | 185 | 26 | 331 | 127 | 10 |
| NRV300 | 300A | 300.0 | 220 | 30 | 370 | 150 | 15 |

# DIMENSIONS OF ACCESSORIES

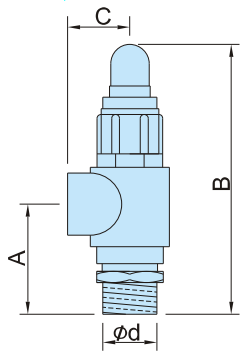
## 90° ELBOW (No.14)




UNIT : mm

| TYPE | BORE | A | B | C | WT.(kg) |
|---|---|---|---|---|---|
| NDE50 | 40A.50A | 95.3 | 120 | 155 | 4-Φ19 |
| NDE65 | 65A | 95.3 | 140 | 175 | 4-Φ19 |
| NDE80 | 80A | 114.3 | 150 | 185 | 8-Φ19 |
| NDE100 | 100A | 152.4 | 175 | 210 | 8-Φ19 |
| NDE125 | 125A | 190.5 | 210 | 250 | 8-Φ19 |
| NDE150 | 150A | 228.6 | 240 | 280 | 8-Φ19 |
| NDE200 | 200A | 304.8 | 290 | 330 | 12-Φ19 |
| NDE250 | 250A | 381.0 | 355 | 400 | 12-Φ19 |
| NDE300 | 300A | 457.2 | 400 | 445 | 12-Φ19 |

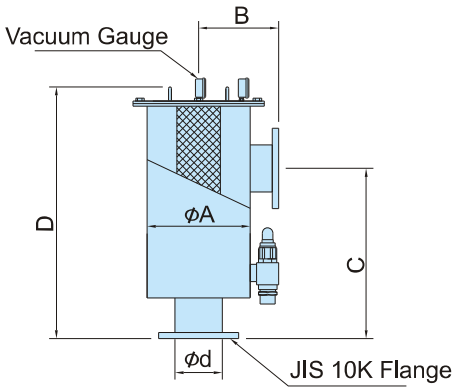
## VACUUM SAFETY VALVE (No.15)




UNIT : mm

| TYPE | A | B | C | Φd | E | Wt (kg) | FITTING MODEL |
|---|---|---|---|---|---|---|---|
| VSV 32 | 83 | 208 | 51 | 32 | PT 1 1/4" | 1.87 | RLV,RWV -40,50,65,80 |
| VSV 50 | 103 | 266 | 67 | 50 | PT 2" | 3.91 | RLV,RWV-100,125,150,200,250,300 |

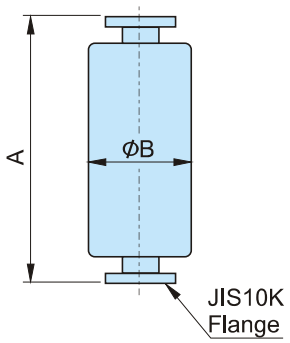
## SUCTION FILTER TANK (No.16)
## -- vacuum gauge included




UNIT : mm

| TYPE | BORE | A | B | C | D | Φd |
|---|---|---|---|---|---|---|
| SFT50 | 50A | 244 | 222 | 340 | 530 | 60 |
| SFT65 | 65A | 244 | 222 | 440 | 630 | 76 |
| SFT80 | 80A | 244 | 222 | 525 | 730 | 89 |
| SFT100 | 100A | 285 | 240 | 525 | 730 | 114 |
| SFT125 | 125A | 360 | 280 | 505 | 730 | 139 |
| SFT150 | 150A | 360 | 280 | 595 | 880 | 165 |
| SFT200 | 200A | 470 | 335 | 650 | 1000 | 216 |
| SFT250 | 250A | 470 | 335 | 800 | 1200 | 267 |
| SFT300 | 300A | 600 | 405 | 1000 | 1400 | 319 |

## DISCHARGE SILENCER (No.18)



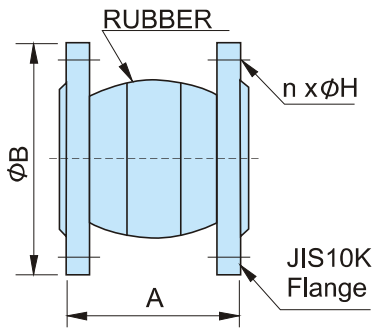

UNIT : mm

| TYPE | BORE | A | B | WT.(kg) |
|---|---|---|---|---|
| DSF50 | 40A.50A | 560 | 140 | 12 |
| DSF65 | 65A | 640 | 165 | 16 |
| DSF80 | 80A | 850 | 210 | 20 |
| DSF100 | 100A | 950 | 240 | 28 |
| DSF125 | 125A | 1150 | 280 | 33 |
| DSF150 | 150A | 1350 | 300 | 41 |
| DSF200 | 200A | 1600 | 400 | 90 |
| DSF250 | 250A | 1700 | 400 | 120 |
| DSF300 | 300A | 1800 | 500 | 140 |

# DIMENSIONS OF ACCESSORIES

## EXPANSION JOINT (No.17)

### EJA TYPE - RUBBER

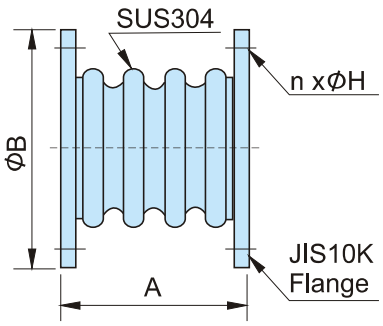



UNIT : mm

| TYPE | BORE | A | B | n-ΦH | EJA WT.(kg) |
|---|---|---|---|---|---|
| EJA50 | 40A.50A | 85 | 155 | 4-Φ19 | 4 |
| EJA65 | 65A | 95 | 175 | 4-Φ19 | 5 |
| EJA80 | 80A | 110 | 185 | 8-Φ19 | 6 |
| EJA100 | 100A | 115 | 210 | 8-Φ19 | 6 |
| EJA125 | 125A | 140 | 250 | 8-Φ19 | 10 |
| EJA150 | 150A | 150 | 280 | 8-Φ19 | 13 |
| EJA200 | 200A | 170 | 330 | 12-Φ 19 | 16 |
| EJA250 | 250A | 210 | 400 | 12-Φ19 | 18 |
| EJA300 | 300A | 240 | 445 | 16-Φ19 | 21 |

### EJB TYPE - SUS 304

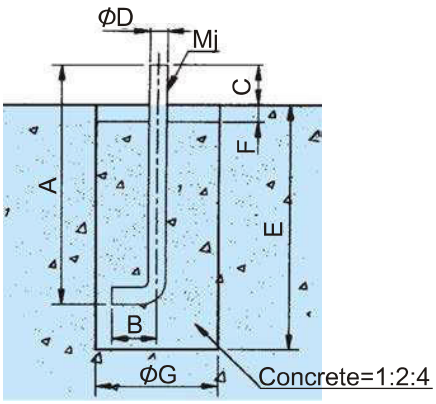
THE EJB TYPE IS RECOMMENDED FOR PRESSURE OVER 5000 mmAq




UNIT : mm

| TYPE | BORE | A | B | n-ΦH | EJB WT.(kg) |
|---|---|---|---|---|---|
| EJB50 | 40A.50A | 85 | 155 | 4-Φ19 | 4 |
| EJB65 | 65A | 95 | 175 | 4-Φ19 | 5 |
| EJB80 | 80A | 110 | 185 | 8-Φ19 | 6 |
| EJB100 | 100A | 115 | 210 | 8-Φ19 | 6 |
| EJB125 | 125A | 140 | 250 | 8-Φ19 | 10 |
| EJB150 | 150A | 150 | 280 | 8-Φ19 | 13 |
| EJB200 | 200A | 170 | 330 | 12-Φ 19 | 16 |
| EJB250 | 250A | 210 | 400 | 12-Φ19 | 18 |
| EJB300 | 300A | 240 | 445 | 16-Φ19 | 21 |

## ANCHOR BOLT (No.19)




UNIT : mm

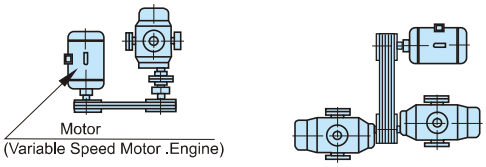
| TYPE | Mj | A | B | C | D | E | F | G | FITTING TYPE |
|---|---|---|---|---|---|---|---|---|---|
| AB1 | M12 | 165 | 30 | 25 | 12 | 165 | 5 | 100 | RL, RW-50,65,80 |
| AB2 | M16 | 245 | 45 | 40 | 16 | 245 | 5 | 120 | RL, RW-100,125 |
| AB3 | M20 | 325 | 55 | 55 | 20 | 325 | 5 | 150 | RL, RW-150,200,250,300 |

perotac
MADE IN TAIWAN

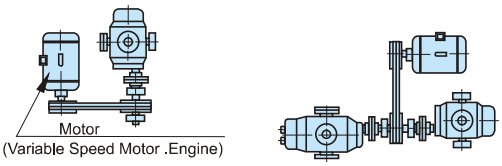
## DRIVING METHOD

Motor and V-belt drive is a standard. Blowers with direct coupling to the motor, engine driven blowers with decelerator, or with speed change gear driving may also be adopted. For power in excess of 90 KW(120 HP), the V-belt driving with a counter shaft becomes a standard.

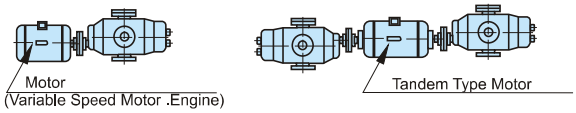
### V-belt Driven



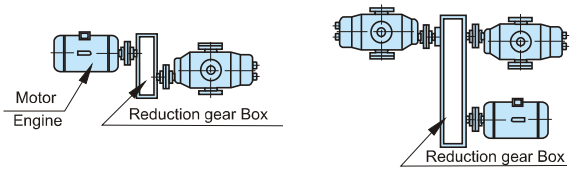

### V-belt Counter Shaft Driven




### Direct Coupling




### Direct Coupling with Reduction Gear Box




## APPLICATION GUIDE

The following applications are typical of many uses for PEROTAC blowers. This is a general listing and does not include all type of applications.

### PRESSURE OPERATION

■ Combustion air for burners, gas burners,coke-fired forges, small foundry cupolas in:

- iron & steel industries
- food processing
- chemical plants
- non-ferrous metal refineries
- smelters & foundries
- bakeries
- glass, ceramics, brick
- oil, tar & asphalt heaters
- on road machinery
- pipe, wrapping machines

■ Drying:

- air and sand blasting
- grain conditioning in elevator storage
- supercharging
- dusting and spraying
- washing, drying and cleaning drums
- blending and mixing
- sand filter regeneration
- dust blowing
- cake blowing
- pressurizing paper mill headboxes

■ Aerating and agitating liquids in :

- waste treatment plants
- waste treatment
- electrotyping & engraving
- oyster washing
- oil blending
- ice manufacture
- chemical processes
- preventing ice formation in power plants
- fish ponds, lakes
- flotation
- blending vinegars
- asphalt refining
- food washing

■ Pneumatic conveying:

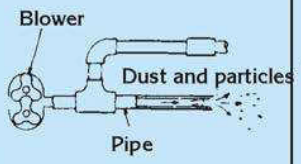
- granular materials
- insulation blowing
- liquid transfer
- fluidizing
- pulp conveying
- wood and metal chip conveying
- rock dusting
- flour and powder

### VACUUM OPERATION

- industrial and commercial vacuum cleaning
- paper folding machinery
- garment presses
- bottle and tube filling machines
- drying
- mat formers
- pump priming
- dust collecting

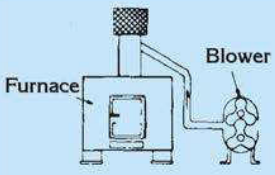
## FOR PRESSURE OPERATION

### PIPE CLEANING



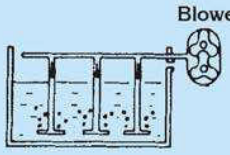

Used to remove dust and rust when replacing piping or conducting periodic inspections.

### FURNACE



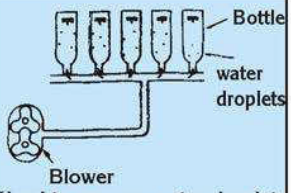

The blower can help boost burn performance and promote exhaust gas removal.

### PURIFICATION TANK



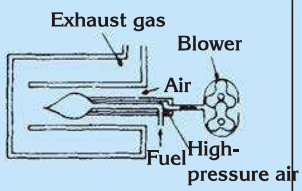

Used to agitate sediment and for purification at water treatment plants.

### BOTTLE WASHER



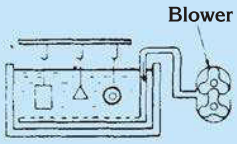

Used to remove water droplets remaining inside bottles after washing.

### GAS BURNER



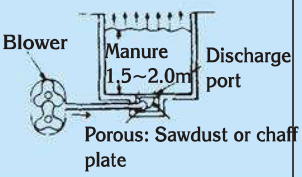

High-pressure discharge air can be used to disperse the fuel.

### PLATING TANK



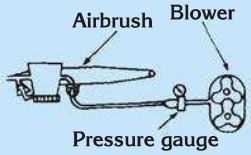

To improve the quality of plating, air is fed into the tank to recirculate the electrolyte. The Roots blower is used as the air source.

### FERTILIZER PRODUCTION USING MANURE FERMENTATION



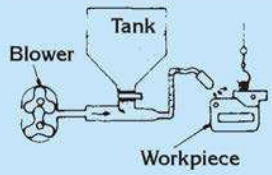

Used to hasten fermentation by feeding air.

### AIRBRUSH



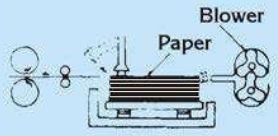

High-pressure discharge air containing no oil is extremely well-received for airbrush applications.

### SANDBLASTING



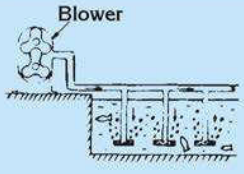

Used as the air source for sandblasting.

### PRINTING PRESS PAPER FEED



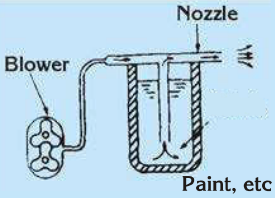

Operations are simplified by using discharge air to separate, align, and feed the paper.

### FISHERY POND ENZYME SUPPLY



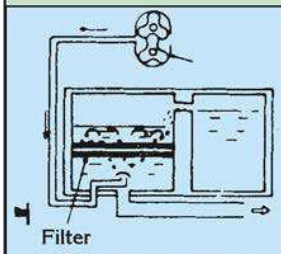

Used for underwater enzyme supply in comparatively shallow ponds.

### SPRAYING



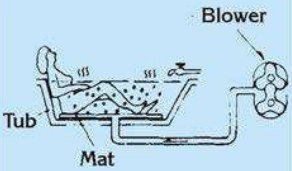

Used to supply high-pressure discharge air containing no oil to the sprayer.

### BACKWASH



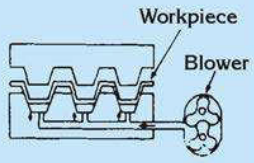

Used as a backwash for filters.

### THERAPEUTIC BATH



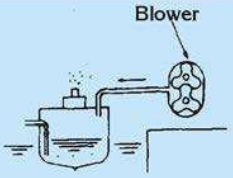

Widely used in hospitals and hotels for healthy bath aeration.

### STAMPING DIE




Used to remove the stamped workpiece from the die.

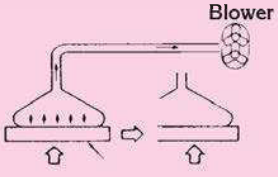
### BALLAST PUMP




Used for liquid chemical ship lowering and other types of ballast pumping.

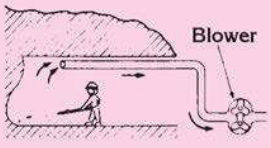
## FOR VACUUM OPERATION

### ABSORBING CONVEYANCE



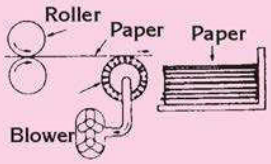

It can convey relatively light materials by absorbing them.

### CONSTRUCTION SITE



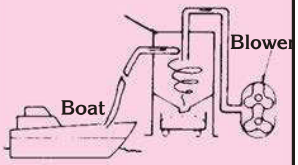

Can be used partial disposal of powders or poisonous gases.

### CONVEYING PAPER




Conveying paper in printing machines.

### PARTICLES TRANSPORT




Used for air transfer of pellet material.